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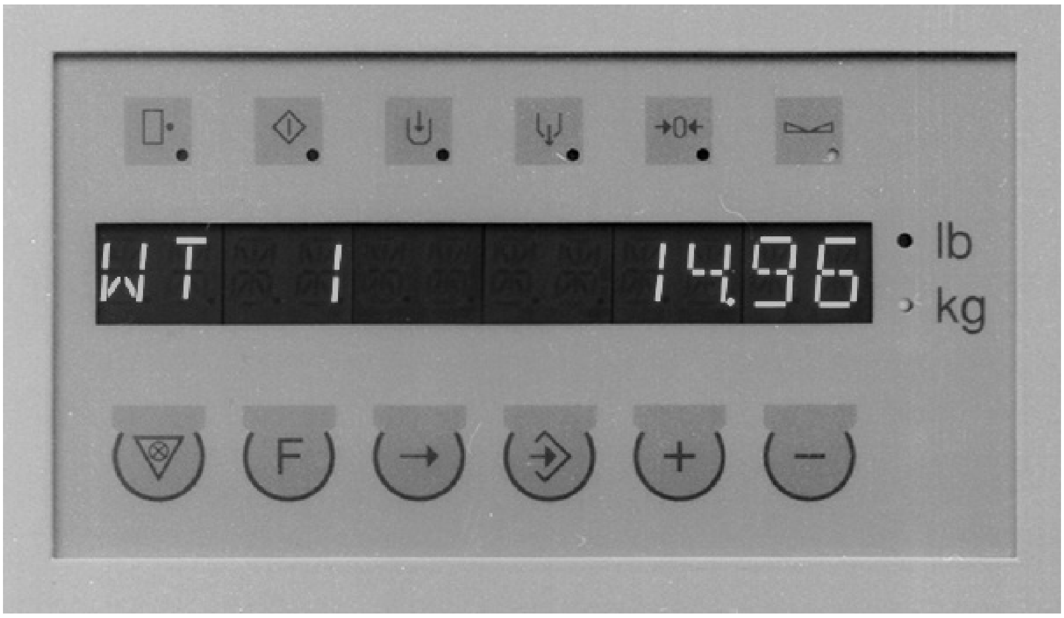
Operating instructions

Scale control unit

Dump scales *(DUMP)*
Differential dosing scales *(DIFFG, DIFF, DIFFM)*
Bagging scales *(BAG)*



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Contents

These operating instructions include basic information on the design and operating principle, the installation, start-up, operation and maintenance of the control unit. All appendices are integral parts of the Operating Instructions.

Safety

This control unit is built in compliance with the recognized safety engineering principles. In spite of this, inexpert use may entail hazards to the health and life of persons, or may cause damage to property.

Warranty

Non-compliance with the Operating Instructions will result in the lapse of warranty; this also applies to alterations or repairs to the control unit without permission in writing from Bühler in advance. Damage which is the result of inexpert handling, disregard of our instructions, or operating errors made by untrained personnel, can under no circumstances be charged to the manufacturer.

Liability

Bühler is only liable for direct injury to persons and direct property damage based on the applicable product liability law if the control unit is used within the area specified in these operating instructions, or in a contractually agreed application. Bühler is not obliged to reimburse damage not directly affecting the control unit itself *(exploitation losses, production downtimes and profit drop as well as other direct and indirect damage)*.

Note:
*These operating instructions apply from program version **SCALE39J.***

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1 Safety

1.1 General

Note:
These operating instructions must always be kept ready to hand near the control unit.

1.1.1 Duty of instruction

The user is responsible for the training and safety of operating personnel. Thus, it is very important that the documentation is actually handed over to these persons.

The operator and operating personnel of the control unit are obliged to note and to observe the instructions in these operating instructions.

Every person who is involved in the connection, operation, or maintenance of the control unit must have read and understood the operating instructions.

The control unit should only be operated by persons who have been instructed and are aware of the dangers involved.

1.1.2 Accident prevention

The control unit is equipped ex factory with safety devices. When operated in accordance with the intended purpose, these devices comply with the actually valid international safety standards and the relevant accident prevention rules.

These operating instructions include the safety rules for preventing accidents. In addition, all generally valid local safety instructions and safety rules do always apply.

Note:
Safety devices, warning and instruction signs must be kept clean and never be removed or covered.
Immediately replace damaged warning and instruction signs with new ones.

1.2 Notes on safety at work

1.2.1 Commissioning and installation

The startup, trial run and adjustment work should only be carried out by instructed and *(if necessary)* authorized personnel.
Prior to the initial startup of the machine, the operating personnel should familiarize themselves with all instructions and regulations contained in these operating instructions.

Generally the country specific standards are to be observed.

1.2.2 Operation of the control unit

Work on the control unit may only be carried out by authorized personnel who have familiarized themselves with all display and operating elements.

1.2.3 Electrical equipment

All electrical installation and inspection work on the machine should be carried out by local authorized personnel. Electrical installations and appliances must be checked at regular intervals.
Pay special attention to the following:

- For all set-up and repair work on the control unit, the power supply has to be interrupted by disconnecting all poles. The mains has to be secured against switching on.
- Immediately repair or replace any defective installations or devices.
- Do not lay loose cables on floors.

1.3 Safety concept for control units

The control systems supplied by Buhler are an integrated part of the safety concept for preventing accidents involving our machines and plants. With respect to the operator, Buhler does not accept liability for malfunctions of control units and damage resulting from such malfunctions.

2 Technical data

Housing	
Dimensions	300 x 380 x 155 mm *(H x B x T)*
Degree of protection	IP65
Weight	approx. 10 kg
Ambient conditions	
Temperature *(in operation)*	–10…+50 °C *(calibratable –10…+40 °C)*
Temperature heatable version *(in operation)*	–30…+50 °C *(calibratable –30…+40 °C)*
Temperature *(device switched off)*	–10…+75 °C
Humidity	max. 95 % no condensation
Power supply	
24 V DC version	according to DIN19240 24 V DC *(+20 % / –15 %)*
Mains version	100…240 V AC *(+10 % / –15 %)* 43…67 Hz
Power consumption	max. 40 VA *(DIFFG with motor max. 200 VA)*
Data backup	at least 20 years *(battery buffered RAM)*
24 V inputs [8/14]	Current consumption max. 15 mA
24 V outputs [8/20]	Load maximum 200 mA short-circuit proof. The total load of the 24 V exerted by external users may not exceed 700 mA.
Analog output 0/4…20 mA [1/2]	max. load resistance 500 Ω *(resolution 800 parts)*
Analog input 0/4…20 mA [0/1]	Input resistance 150 Ω *(resolution 500 parts)*
Serial interface RS-485 [1/2] *(RS-422 possible)*	Bühler standard scale protocol for remote system *(according to manual 66435)* or remote control according to appendix
Serial interface 20 mA [0/1]	Printer connection according to appendix
Force transducer *(DMS)*	
Bridge feed	10 V
Input measuring range	0,5…23 mV
Internal resolution	approx. 900’000 parts for input measuring range
Calibratable resolution	max. 3000 d
Load cell resistance	58…400 Ω max. 6 load cells with 350 Ω *(calibratable 87…400 Ω, 4 load cells with 350 Ω)*
Min. permissible input signal	2 µV / calibration value *(calibratable)*
Noise emission	EN 50081-1 class B fulfilled
Noise immunity	EN 50082-2 industrial range satisfied

[x/x] = [number of IO on basic print / number of IO with option IO extension]

3 Common features of all scale types

Note:
The present manual applies for all different scale types. This chapter deals with the common features of all types. The type-specific features are dealt with in the following main chapters.

3.1 Overview of scale types and versions

The scale control unit MEAF is a universal scale electronics which is installed to standard on the scale. The MEAF contains the control programs for **dump scale, bagging scale** and **differential dosing scale.** All important functions of the previous scale control units MWEE, MWEB, MWEG, MWEO, MEAE, MWET and MEAP are included. The settings are done on the display in the front door *(adjustment and parameterization)*. The scale is operated in local mode on the display, in remote mode via the serial or analog interface.

Scale type	Scale name	Parameters SYS.TYP [1])
Batching scale	MWBL, MWBB, MWBS, MSDL, MSDT	TYPE = **DUMP**
Bagging scale	MWBB, MWBW	TYPE = **BAG**
Differential batching scale with feed gate	MWBG, MSDG	TYPE = **DIFFG**
Differential batching scale with screw feeder	MWBO, MSDA	TYPE = **DIFF**
Micro-differential batching scale with screw feeder	MSDE, MSDF	TYPE = **DIFFM**

[1]) See chapter «Operation and general parameters»

3.2 Basic design versions

The following basic design versions of the MEAF scale control unit are available:

- Power supply 24 V DC or 115/230 V AC
- Non-calibratable or calibratable
- Painted steel housing or stainless steel housing
- Heatable version for installation in very cold surroundings

The complete list of the available versions and the order numbers can be found in the chapter 7.3 «Spare parts».

Note:
The calibratable versions contain the option IO extension and a sealable cover for the calibration switch. These versions are factory-adjusted according to the calibration requirements.

3.3 Options

For order numbers please refer to chapter 7.4.

3.3.1 IO extension

Plug-in print

This option can also be installed later at any time. This option is already installed in the calibratable versions. The option is required in the following cases:

- Analog flow rate setting 0/4…20 mA
- Admixture mode in differential batching scale
- Additional digital inputs or outputs 24 V
- Connection of a printer 20 mA CL
- Second interface RS-485 for host or remote control
- Second analog output 0/4…20 mA for actual flow rate with DIFF and DIFFM

3.3.2 Remote control

Additionally to the local operation mode; see also appendix
Built-in version to be installed into the cabinet door

Table or wall-mounted version

3.3.3 DMS connection box

For the extension of the force transducer cables; see also appendix

- with 10 m extension cable
- without cable *(only box)*

3.3.4 Printer

For types, connection and order numbers see appendix.

3.3.5 Profibus DP

Plug-in print

The Profibus extension print can also be installed at a later stage.

Also see manual 66499 Profibus DP protocol MEAF.

3.4 Structure of the control unit

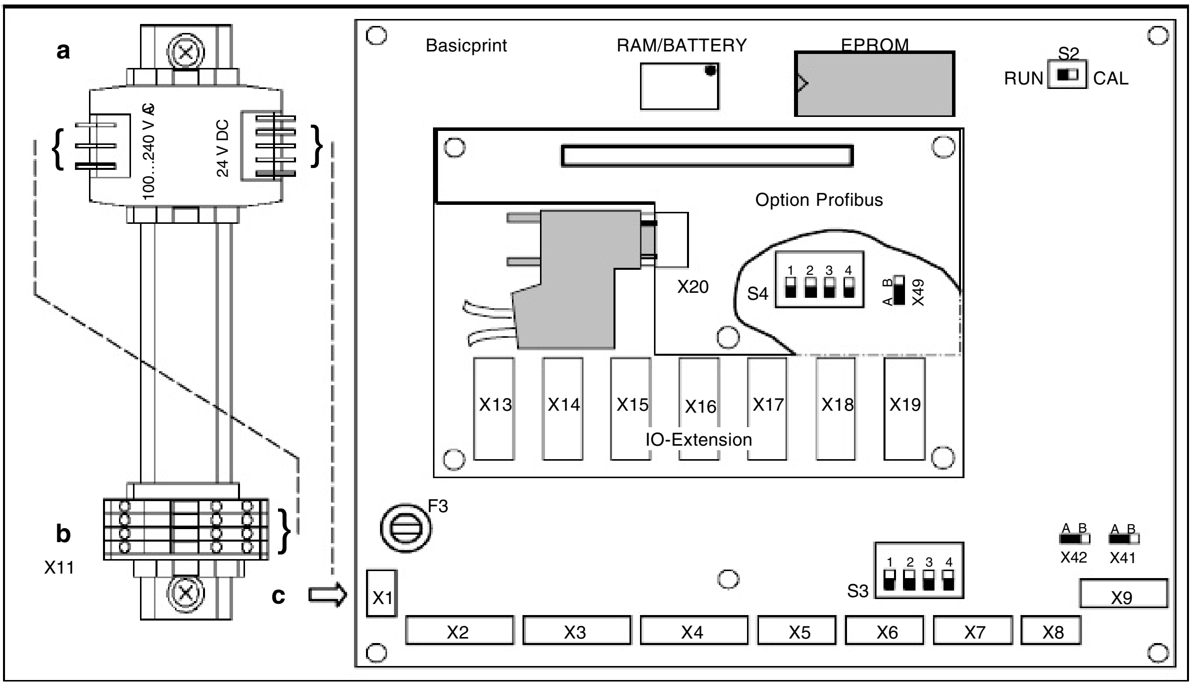


Fig. 3.1

a) Mains supply (AC/DC converter)

b) Mains voltage connection 100…240 V AC to X11
see Appendix on mains supply for wiring details

c) In case of external +24 V DC power supply connection is made directly to X1

Mains supply Only in the 115/230 V AC version

IO extension Option; plugged in and screwed onto the basic print

Profibus DP Option; plugged in and screwed onto the basic print or IO extension

EPROM Program module. To change the program the EPROM must be lifted carefully with a screwdriver.

Caution!

The notch must be on the left!

S2 Calibration switch. RUN = release weighing operation / CAL = calibration

S3, S4 Settings for RS-485 host interface. All switches are OFF in standard position. If the device is the last one on the bus, the switch S3:3 or S4:3 must be set to ON. Possibly the switches S3:1 and S3:2 at the last device must be set to ON *(or S4:1 and S4:2).* These are bias resistors which are normally available at the Host. *(For details see manual 66435)*

X41, X42 Plug-in jumpers of the DMS input *(sense lines)*
Position A = 4-wire DMS *(standard)* / position B = 6-wire DMS

X49 Position A = analog input 0/4..20mA at terminals AI+ / AI–
Position B = calibratable *(AI used internally for temperature measurement)*
(plug-in jumper only available from print version -02)

F3 Fine fuse 1.25 A slow-blow as 24 V input fuse

3.5 Connection of power supply

Danger!

The protective cover in the MEAF scale electronics may only be removed if the power supply is interrupted at all poles.

Power supply

Depending on the version, the MEAF is supplied with 24 V DC or with a single-phase mains voltage:

Version 24 V DC		*(see name plate)*
• 24 V	X1:1	*(no connection here with version 115/230 V AC)*
• 0 V	X1:2	

Version 115/230 V AC	*(see name plate)*	
• Phase	X11:1	(L1)
• Protective earth	X11:2	(PE)
• Neutral conductor	X11:3	(N)

Shield

Shielded cables must be used for the **serial interface** and the **analog current signals.** The cable shields must be connected as short as possible to the terminals marked «Shield».

3.6 Commissioning and adjustment of the scales

3.6.1 Basic procedure for every commissioning

Switch S2, see chapter «3.4»

- Remove transportation locks of the weigh hopper.
- Check whether the weigh hopper is suspended freely.
- Before the power supply is switched on, the protective cover must be mounted and the cover closed.
- After the scale electronics have been switched on for at least 10 minutes, the scales can be adjusted according to the following chapter «Adjustment of the scales».
- If the fault 06 MLOST is displayed during start-up then the cause for that has to be eliminated first *(see chapter «3.8»)*.
- In addition the most important parameters must be checked or set and the commissioning carried out with product. Here the procedure must follow the instructions in the chapter «Commissioning» of the respective scale type.

3.6.2 Adjustment of the scale

For parameters see chapter «Operation and general parameters» .

- The scale must be completely emptied. «F» key for parameter ADC. **DISC**.
- The calibration switch S2 must be set on the basic print to « **CAL**».
- Activate **zeroing** by pressing the «F» key for parameter ADC.**ZERO**. «SET» flashes during the zeroing.
- At parameter ADC.**CALW**, the total weight of the used **calibration weights** must be entered. *(As a guideline the standard value of the calibration weight should be approx. 50 - 100% of the loading capacity.)*
- The calibration weights must be mounted symmetrically on the weigh hopper at the places provided.
- Trigger **calibration** by pressing the «F» key at the parameter ADC.**CALW**. «SET» flashes during calibration.
- Check weight display for the calibration weight at parameter ADC.WT.
- Remove calibration weights.
- Check weight display (0) at parameter ADC.WT.
- The calibration switch S2 must be set to the «RUN» position.

3.7 Operation and general parameters

3.7.1 Display operation

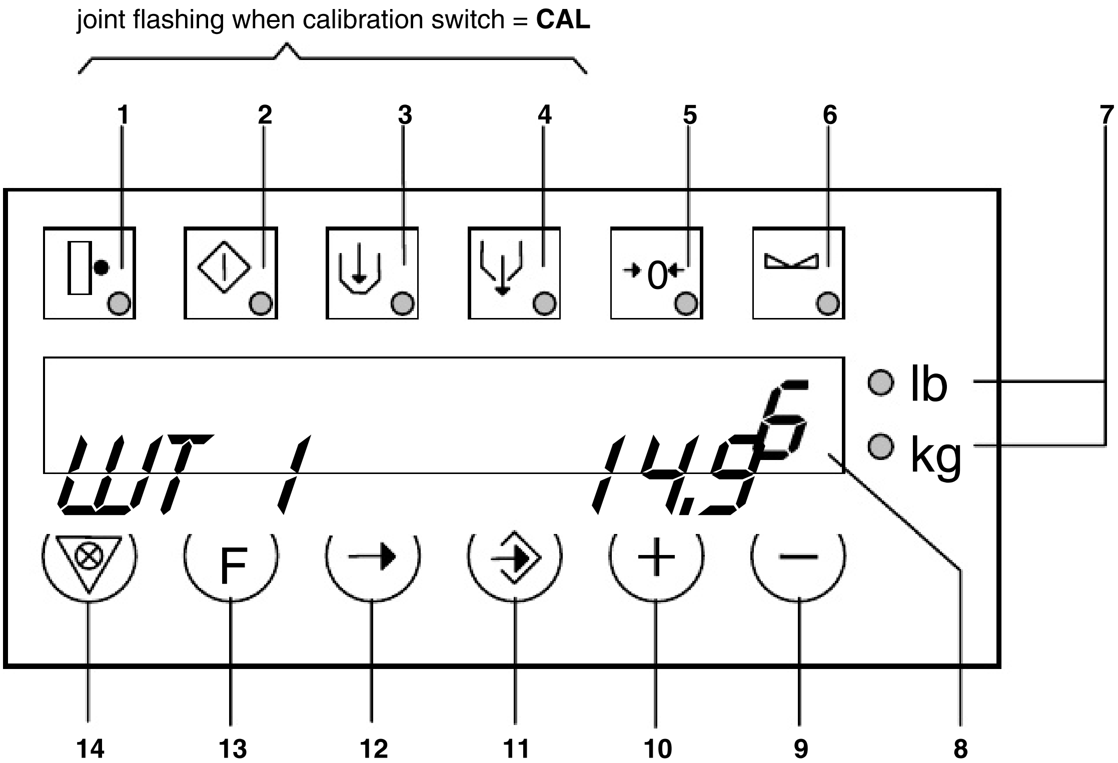


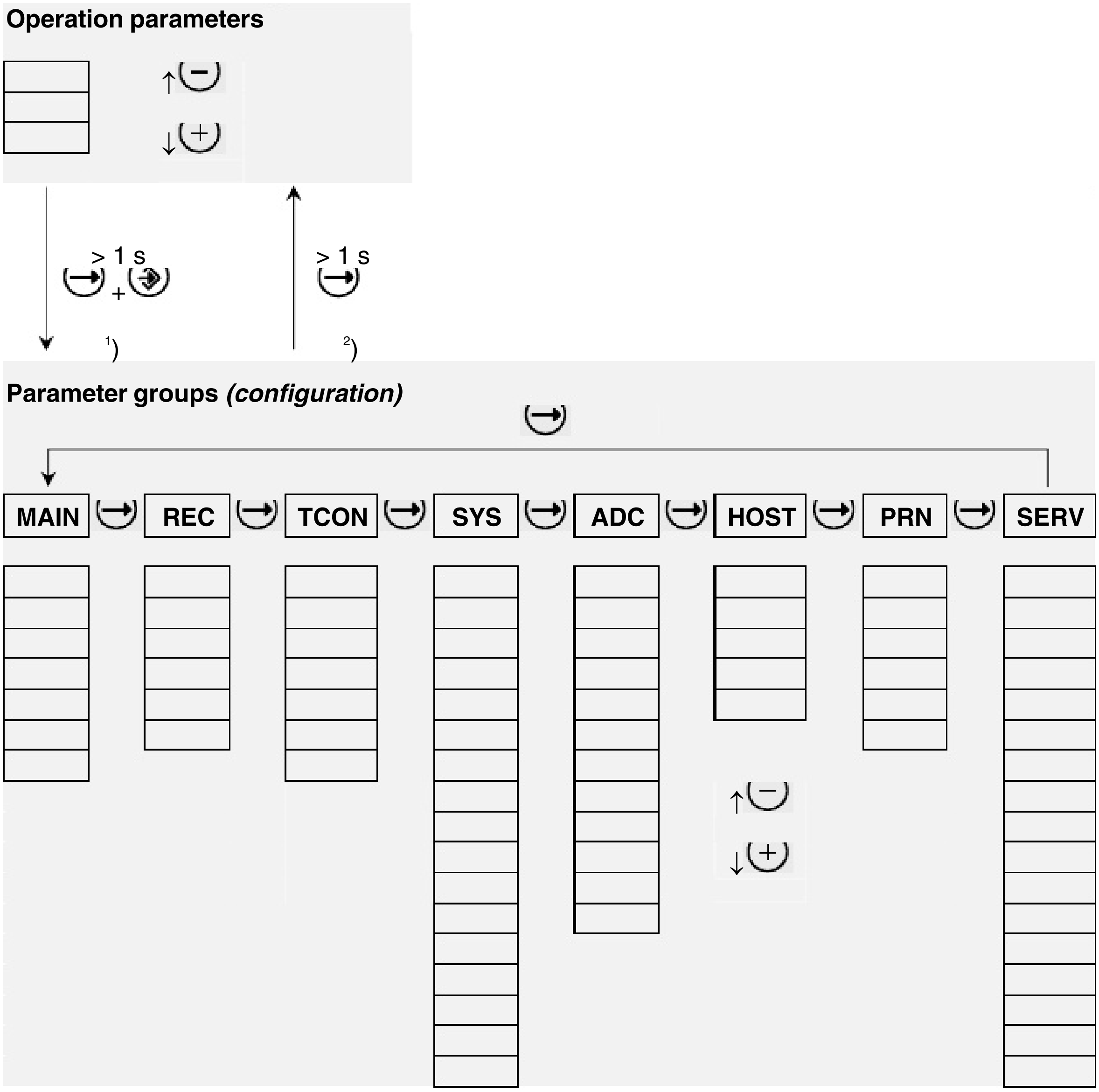
Fig. 3.2

1	*Remote (LED)*	
2	*Weighing process started (LED)*	
3	*Filling/Dosing (LED)*	
4	*Emptying/Dosing (LED)*	
5	*Weight in zero range (LED)*	
6	*Scale standstill (LED)*	
7	*Weight unit in lb or kg (LED)*	
8	*Display of parameters with code and value*	
9	*«Minus» key*	*previous parameter*
10	*«Plus» key*	*next parameter*
11	*«Enter» key*	*data input*
12	*«Arrow» key*	*next parameter group*
13	*«F» key*	*special functions depending on parameter*
14	*Lamp test key and clear alarm*	

3.7.2 Alter parameters

1. Press > 1 s *(press longer than 1 sec. until decimal point is flashing)*
2. , increases respectively reduces the value or select the decimal place to be altered directly with
3. Press briefly *(until decimal point is not flashing anymore)*

3.7.3 Select parameters



1) Change Operation parameters → Parameter groups press and together for longer than 1 sec.

2) Change Parameter groups → Operation parameters press for longer than 1 sec.

3.7.4 Overview of general parameters

The groups SYS, ADC, HOST, PRN and SERV are valid for all scale types. Standard operation parameters and the groups MAIN, REC and TCON are described in the chapters of the scale types.

MAIN	TCON	SYS System configuration	ADC Analog-Digital Converter	HOST Serial host interface	PRN Printer	SERV Service
		TYP Scale type	WT Current scale weight	ADR Address	PEVT Print events	VER Program version
		REM Local/Remote	DISC Discharge completely	BAUD Baud rate	PCYC Print cycle data	DATE Date
		RIN Remote input	ZERO Zeroing	TOUT Timeout	PJOB Print job report	TIME Time
		EIN Enable input	CALW Calibration weight	WFOR Weight format	PLEN Length of page	ERR 3 last errors
		DISP Remote control	MAXR Measuring range	ENBL Start/stop serial	LFTM Left margin	CPU Load
		ACLR Clear alarm	DIV Division		MJOB Several jobs	SADR System address
		AINV Invert alarm output	UNIT Unit		PALA Printer alarm	PCNT Number power-up
		WIMP Weight pulse	FILT Filtering		JOBL Length job print.	DMS DMS input signal
		IMPM Mode weight pulse	MOT Standstill tolerance		DPOS Pos. date/time	+10V DMS bridge voltage
		AO Analog output	SENS Line compensation		NPOS Pos. total No.	AOUT Analog output
		AO2 Analog output 2	AZER Automatic zeroing		SPOS Pos. total weight	AOU2 Analog output 2
		AI Analog input	NORM Calibratability		S2PO Pos. undeletable total weight	AIN Analog input
		DI1…DI8/14 Digital input	TCMP Temp. comp.			DIN Digital input 1…8
		DO1…DO8/20 Digital output	ZADV AD value Zero			DIN9 Digital input 9…14
		SIO Serial interface	DADV AD value Calw			DO Digital output 1…8
		SIO2 Serial interface. 2				DO9 Digital output 9…16
		SIO3 Serial interface. 3				DO17 Dig. output 17…20
		INIT Basic initialization				REL Relay
						SIO Serial interface
						SIO2 Serial interface 2
						SIO3 Serial interface 3
						P-DP Profibus DP
						PARP Parameter output

3.7.5 Description of the general parameters

Code	Parameter group SYS *(system configuration)*	Input range
TYP	Scale type 2) ----- = No scale selected, all outputs are without signal *(e.g. after loss of data)* DUMP = Dump scale BAG = Bagging scale DIFFG = Differential dosing scale with feed gate DIFF = Differential dosing scale with screw feeder DIFFM = Micro differential dosing scale with screw feeder DMST = DMS transmitter *(manual 66388)* ICONV = Pulse converter *(replacement MWEX, manual 66443-1)* FBAL = Automatic volume controller and flowmeter MZAH *(manual 66469-1)* MZMN = Micro dosing unit *(manual 66488-1)* CHECK = Check weigher for bagging lines *(manual 66476-1)* SLIDE = Metering slide gate *(manual 66548-1)* DCOS = Dosing control for batching scale or loading *(**Caution:** diverse parameters are initialized when switching over)*	-----...DCOS [-----]
REM	Local/remote mode LOC = local *(operation on the display)* REMS = remote – serial *(operation via serial interface)* REMA = remote – analog *(nominal rate 0/4…20 mA)* REMP = remote – Profibus DP, operation exclusively via Profibus *(setpoint assignment and control signals via Profibus active)*	LOC…REMP [LOC]
RIN	24 V input remote OFF = remote input 0 V or 24 V -> mode acc. to REM ON ≡ remote input ≡ 0 V -> mode ≡ LOC ≡ remote input ≡ 24 V -> mode ≡ acc. to REM	OFF/ON [OFF]
EIN	24 V input enable OFF = input has no function ON = input is always in function ONLOC = input is only in function in local mode ONREM = input is only in function in remote mode	OFF…ONREM [OFF]
DISP	Remote control LOC = operation local *(remote control display only)* REM = operation at remote control *(local display only)* BOTH = operation local and at remote control	LOC…BOTH [LOC]
ACLR	Clear alarm OFF = the alarms are acknowledged automatically ON = the alarms are acknowledged only with the key, the serial interface, Profibus or the 24 V input «Clear alarm».	OFF/ON [OFF]
AINV	Invert alarm output OFF = 24 V at alarm ON = 0 V at alarm	OFF/ON [ON]
WIMP	Weight pulse in kg *(weight per pulse)* 4) 3) With IMPM = TIME the value must be selected so that approx. 2 pulses are transmitted per dumping or cycle.	1…99999 [10.00]

2) Only adjustable or executable if the calibration switch is in the «CAL» position

3) Not available in the scale type BAG

4) Unit according to UNIT, decimal places according to DIV

[ ] Standard setting

Code	Parameter group SYS *(system configuration)* *(continued)*	Input range
IMPM	Mode weight pulse 3) OFF = weight pulse switched off PACK01 = pulse packet with 0.1 s pulse and pause PACK05 = pulse packet with 0.5 s pulse and pause TIME = pulses distributed chronologically with 0.5 sec. pulse	OFF…TIME [TIME]
AO	Analog output switching 0/4…20 mA	0MA/4MA [0MA]
AO2	Analog output 2 switching 0/4…20 mA 1)	0MA/4MA [0MA]
AI	Analog input switching 0/4…20 mA 1) «F» key: Print temperature in °C if X49 = B	0MA/4MA [0MA]
DIx	Signal assignment digital inputs 24 V 2) 5) *(Selection with «F» key and simultaneously plus/minus key)*	0…20
DOx	Signal assignment digital outputs 24 V 2) 5) *(Selection with «F» key and simultaneously plus/minus key)*	0…20
SIO	Application of serial interface RS-485 basic print 2) OFF not used HOST Bühler standard scale protocol DISP remote control REMD remote display 6-digits *(only DMS-T)* RS422 Bühler standard scale protocol with RS422, SIO is sender *(possible only with option IO extension)* FIN2 2 frequency converters *(only DIFFM; without mixer)* FIN3 3 frequency converters *(only DIFFM; with mixer)*	OFF…FIN3 [HOST]
SIO2	Application of serial interface 2 RS-485 1) 2) OFF not used HOST Bühler standard scale protocol DISP remote control REMD remote display 6-digits *(only DMS-T)* RS422 Bühler standard scale protocol with RS422, SIO2 is receiver	OFF…RS422 [HOST]
SIO3	Application of serial interface 3 current loop 1) 2) OFF not used PRNO printer OKI Microline ML280 or ML3320 PRNE printer EPSON LQ570 FDW PRNX printer with X-ON / X-OFF protocol PRNN printer without protocol *(2-wires no acknowledgement from the printer)* REMD remote display 6-digits *(only DMS-T)*	OFF…REMD [PRNO]
INIT	Basic initialization of all parameters with the «F» key. 2) «URINIT» is displayed during basic initialization afterwards only the parameter SYS.TYP. A valid type has to be selected first in order for the full list of parameters to be available again. *(Total data loss, perform commissioning)*	

1) Only with option IO extension
2) Only adjustable or executable if the calibration switch is set to «CAL»
5) It is advisable not to change the standard setting.
With a loss of data *(program change or replacement print)* the standard setting will get active again!
[ ] Standard setting

Code	Parameter group ADC *(Analog Digital Converter)*	Input range
WT	Current scale weight in kg 2) «F» key: Current scale weight with a 10-fold division	
DISC	Discharge completely with «F» key *(complete emptying of weigh container)* Display of step number and current scale weight	
ZERO	Zeroing with «F» key. «SET» is flashing during zeroing. 1)	
CALW	Calibration weight in kg 2) 1) The input value corresponds to the applied calibration weight. Calibrate by pressing the «F» key. «SET» is displayed during calibration.	100…100000 [40.00]
MAXR	Range monitoring of the weight in kg 2) 1) 0 = range monitoring switched off > 0 = alarm AD converter if the current weight is > MAXR +9 DIV or < –5 % of MAXR	0…999999 [0]
DIV	Division (d) of the weight displays 1) If DIV is adjusted, all weight values are initialized. «WTINT» is displayed on the display.	0.001…50 [0.01]
UNIT	Unit of the weight displays 1) NO = no weight unit KG = weight unit kg *(kg_LED in weight values)* LB = weight unit lb. *(lb.-LED in weight values)* If the weight is displayed in lb., the variables FLO, FMAX, FMIN are displayed in cwt./h *(100 lb./h)*	NO / KG / LB [KG]
FILT	Number of weight values for the digital filtering. 1) If the «F» key is pressed, the internal variables are displayed	2…64 [2]
MOT	Standstill tolerance in number d *(DIV)* 1)	0…9.9 [0.5]
SENS	Line compensation load cell extension cable 1) OFF = no line compensation *(4-wire technique)* plug-in jumpers X41=A, X42=A ON = line compensation *(6- wire technique)* plug-in jumpers X41=B, X42=B ON-02 = line compensation for basic print with index –02 *(old)* plug-in jumpers X41=A, X42=B UIN = Sense-input used as voltage input *(only FBAL or BAG with bed depth adjustment)* If this parameter is altered a new weight calibration has to take place.	OFF/ON/ON-02/UIN [OFF]
AZER	Automatic zeroing, tolerance in kg 2) 1) 0 = the automatic zeroing is switched off > 0 = an automatic zeroing takes place if the current weight value is within the set tolerance and if there is a standstill. In order for the function to operate a value of > 0 has to be set at MAXR. The zeroing range is limited internally in every case to maximal ± 2 % of MAXR.	0…1000 [0.00]

1) The parameters are only adjustable or executable when the calibration switch is set to «CAL»
2) Unit according to UNIT; decimal places according to DIV
[ ] Standard setting

Code	Parameter group ADC *(Analog Digital Converter)* *(continued)*	Input range
NORM	Calibratability 1) INDST = Industrial mode for applications within the plant OIML = Control system for calibratable applications *(brackets with total printout, total delete only in step 1)* NTEP = Control system for calibratable applications in North America *(no brackets with total printout, total delete only in step 1)*	INDST/OIML/NTEP [INDST]
TCMP	Temperature compensation in ppm/°C *(only for calibratable version)* 1) The values of the sticker on the basic print must be entered here. *(left value if SENS=OFF, right value if SENS=ON)* *(100 = no compensation, 80 = –20 ppm/°C, 120 = +20 ppm/°C)*	70…130 [100]
ZADV	Internal zero point AD value *(is stored upon zeroing)* 1)	
DADV	Internal difference AD value between the calibration value and the zero point *(is stored upon calibration)* 1)	

1) The parameters are only adjustable or executable when the calibration switch is set to «CAL»

[ ] Standard setting

Code	Parameter group HOST *(serial interface SIO/2, Profibus DP)*	Input range
ADR	Communication address	0…255 [0]
BAUD	Baud rate of the serial interfaces With SIO2 the max. baud rate is 9600, also if the setting is 19200	4800…19200 [4800]
TOUT	Timeout time in seconds The device stops metering if there is no more communication for the duration of the timeout time. 0 = no timeout monitoring 1…99 = timeout time *The check only takes place with the setting SYS.REM=REMS.* *In the case of Profibus DP the timeout monitoring is automatically started with a set time, the moment communication has taken place once via the Profibus. If the remote mode SYS.REM has been changed then the timeout monitoring is switched off until the next communication via the Profibus interface.*	0…99 [0]
WFOR	Weight format for serial interfaces and Profibus DP FIX = fix format same as old scale control units kg for DUMP, DIFFG and DIFF g for DIFFM according to ADC.DIV for BAG DIV = according to ADC.DIV	FIX/DIV [FIX]
ENBL	Start/stop via the serial interface *(Command 'S','A')*	OFF/ON [OFF]

[ ] Standard setting

Code	Parameter group PRN *(printer SIO3)* 1)	Input range
PEVT	Print events *(start, stop, alarms)*	OFF/ON [OFF]
PCYC	Print cyclic data according to type of scale *(e.g. after every dumping)* OFF = No cyclic printout SHORT = Short printout FULL = Detailed service printout	OFF…FULL [OFF]
PJOB	Print job report Printout with «Clear total' or once the pre-selected number of bags is reached.	OFF/ON [OFF]
PLEN	Page length 0 = No page feed *(e.g. printing on endless paper)*. Every page is filled completely. > 0 = Page feed, if the printed number of lines correspond to the page length *(see also MJOB)*.	0…96 [0]
LFTM	Left margin Number of spaces on the left for all printouts *(staple margin)*	0…14 [0]
MJOB	Several job printouts on the same page 2) *(only if PLEN > 0)* OFF = A job printout is printed each time on a new page. ON = A complete job printout is printed on the current page, if there is still enough space available.	OFF/ON [OFF]
PALA	Printer alarm 2) 3) OFF = Printer alarm not on alarm output *(only display)* ON = Printer alarm on alarm output Total is only erased after a correct printout *(with Dump)*	OFF/ON [OFF]
JOBL	Length of job printout with total delete 2) Adds empty lines at the end of the job printout. For the printing of short forms or labels on endless paper, if PLEN = 0	0..96 [0]
DPOS	Position date/time with job printout *(at left line, at right column)* 2) *(Line = 0 no printout date/time)*	0..96 1..60 [1 1]
NPOS	Position total number SCNT with job printout 2) *(Line = 0 no printout total number)*	0..96 1..60 [1 20]
SPOS	Position total weight Σ1 with job printout *(at left line, at right column)* 2)	1..96 1..60 [1 40]
S2PO	Position undeletable total weight Σ2 with job printout 2) *(line at left, column at right; line = 0 no printout)*	0..96 1..60 [0 1]

1) Only with option IO extension and when SYS.SIO3 is set to PRNO or PRNE
2) Only with DUMP
3) Only adjustable if the calibration switch is set to «CAL»
[ ] Standard setting

Code	Parameter group SERV *(Service information)*	Input range
VER	Program version *(left)* and data version *(right)*	
DAT	Date	1-1-00…31-12-99
TIM	Time *(seconds = 80 -> clock stopped)* In standard operation the clock is switched on. A battery test is carried out every 24 hours only if the clock is switched on.	0-0-0…23-59-59
ERR	The last 3 alarms. Left the latest, right the oldest alarm. **Extended alarm display:** To change to the extended alarm display the «F» key has to be pressed. The last 9 alarms can be displayed by means of the «F» and «Plus» keys or the «F» and «Minus» keys. The date and time when the alarm occurred can be displayed on the relevant alarm with the «Enter» key. After the 9th alarm «CLEAR ALARM» is displayed. All alarms can be cleared there with the «Enter» key.	
CPU	Computer load in %	
SADR	System address for a program error	
PCNT	Counter number of power failures	
DMS	DMS input voltage *(current voltage)* in mV	
+10V	DMS bridge feed in V *(is only refreshed in step 1)*	
AOUT	Analog output *(actual current)* in mA 2) Current adjustable for tests if calibration switch = CAL	
AOU2	Analog output 2 *(actual current)* in mA 1) 2) Current adjustable for tests if calibration switch = CAL	
AIN	Analog input *(actual current)* in mA 1) 2)	
DIN	Digital inputs 1…8 on basic print *(left DI1, right DI8)*	
DIN9	Digital inputs 9…14 on IO extension option *(left DI9, right DI14)* 1)	
DO	Digital outputs 1…8 on basic print *(left DO1, right DO8)*	
DO9	Digital outputs 9…16 on IO extension option *(left DO9, right DO16)* 1)	
DO17	Digital outputs 17…20 on IO extension option *(left DO17, right DO20)* 1)	
REL	Relay outputs *(left CLOSE, right OPEN; only MWBG)*	

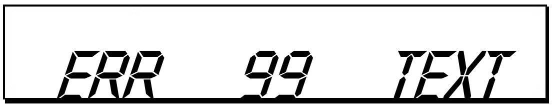
1) Only with option IO extension
2) Trimmable with «F» and (+) or (−) keys if calibration switch = CAL

Code	Parameter group SERV *(Service information)*	Input range
SIO	Serial interface RS-485 to host *(Communication counter, command, error number)* Communication counter: is incremented by 1 per communication with the correct address Command: with lower case letters with a leading «–» Error numbers: 0 = no error 1 = undefined command 2 = wrong checksum in command message 3 = wrong number of characters in command message 5 = wrong character in command message	
SIO2	Serial interface RS-485 *(as SIO)* 1)	
SIO3	Serial interface 20 mA current loop to printer 1) *(communication counter, print job, print status)*	
P-DP	Profibus DP diagnosis 3) 0 = timeout / communication interrupted 1 = error during initialization 2 = no error	
PARP	Activate parameter printout with F key P SIO Parameter output on SIO P SIO2 Parameter output on SIO2 P SIO3 Parameter output on SIO3 P PRN Parameter printer output R SIO Recipe output on SIO R SIO2 Recipe output on SIO2 R SIO3 Recipe output on SIO3 R PRN Recipe printer output	[P PRN]

1) Only with option IO extension
3) Only with option Profibus-DP

3.8 General alarms

In case of an alarm, an alarm number with the appropriate text is displayed alternately with the normal display:



In place of 99 and text, the MEAF alarm number with alarm text appears according to the following table:

Alarm number	Meaning	Remedy
1 ERPOM	**EPROM** An error in the program memory was detected when starting up	Switch MEAF off and back on. 1) If the alarm is repeated, the EPROM must be changed.
2 RAM	**RAM** An error in the data memory was detected when starting up	Switch MEAF off and back on. 1) If the alarm is repeated, the basic print must be changed.
3 RTIME	**Runtime error software** Data have adopted invalid values during the program run	If the alarm is repeated, the EPROM must be changed. Contact Bühler Service and inform them of the system address *(parameter SERV.SADR)* and the program version *(SERV.VER)*. 3)
4 WDOG	**Watch dog** Time run-out in a program cycle	Switch the MEAF off and back on. 1) If the alarm is repeated, the EPROM or the electronics must be changed. Contact Bühler Service.
5 NO24V	**24 V voltage too low**	Check 24 V 2)
6 MLOST	**Data loss** Incorrect data are detected in the data memory *(RAM)* when starting up. The data are initialized automatically to the original values.	– Check the contacts of the buffer battery – Replace battery, located above the memory module *(RAM)*. Afterwards the parameters have to be checked. 3) The scale type *(SYS.TYP)* is set automatically to "-----" *(no scale selected)*. Afterwards it is necessary to first select a valid scale type. **Caution!** A complete commissioning must be performed.

1) Acknowledgement by switching off and back on
2) Automatic acknowledgement when the cause of the alarm is eliminated
3) Acknowledgement with key or 24 V input «Clear alarm»

Alarm-number	Meaning	Remedy
7 RANGE	**AD converter range** Measuring range exceeded or dropped below. See also parameters WT and MAXR	– Repeat calibration 2) – Clear external influence on the weight hopper *(remove transportation locks)* – Check force transducer cable – Change force transducer or basic print
8 ADCAL	**AD converter calibration** Too few AD values are available for calibration	– Repeat calibration 2) – Clear external influence on weight hopper *(remove transportation locks)* – Is force transducer installed in the correct position?
9 ADFCT	**AD converter function** Converter module supplies no AD values	– Change basic print 2)
10 AD10V	**AD converter bridge feed** Load cell voltage supply The bridge feed is outside the permissible tolerance	– Check plug-in jumpers X41, X42 *(sense signals)* 2) – Replace force transducer – Change basic print
11 <4MA	**Nominal value less than 4 mA** For analog nominal value specification 4…20 mA *(remote mode analog)*	– Measure current 2) – Set opposite device to 4…20 mA as well – Eliminate line interruption – Replace IO extension option
12 HOST	**Timeout on host interface** The host system does not communicate anymore with the MEAF electronics	– Start up host 2) – Interface cable interrupted, incorrectly connected – Enlarge HOST.TOUT – HOST.TOUT = 0 *(switch off)*
13 DISP	**Remote control** Serial communication to the remote control is interrupted *(not on alarm output)*	– Check wiring 2) – Set print switch remote control correctly – Remote control connected to the correct terminals according to SYS.SIO or SYS.SIO2
14 P-DP	**Profibus** Profibus error or the communication is interrupted *(timeout)* Alarm only if REM=REMP. Unit stops in case of alarm	– Is the Profibus master active? – Check Profibus wiring and address HOST.ADR ***Note:** This alarm can only occur if a Profibus DP extension card is installed. The exact cause of the error can be read at the parameter SERV.P-DP (Profibus diagnosis).*
16 PRN	**Printer** Printer fault *(Alarm output according to PRN.PALA)*	– Check wiring 3) – Correct printer set at SYS.SIO3 – Printer not selected – No paper
18 BAT	**Battery Low** The voltage of the buffer battery is insufficient	– Check the contacts of the buffer battery 4) – Replace battery, located above the memory module *(RAM)*. Afterwards the parameters have to be checked.

2) Automatic acknowledgement when the cause of the alarm is eliminated
3) Acknowledgement with key or 24 V input «Clear alarm»
4) The battery test is only active if the latest MEAF print version with RAM *(battery)* – base M48T58 is available

4 Dump scale *(DUMP)*

4.1 Description

Operating mode continuous scale: *(receiving scale, check scale)*

The scale registers the continuous product. With rest discharging, a rest dump is added to the total. *(Nominal rate = 0 / nominal total weight = 0)*

Operating mode rate preselection: *(mill feed B1 scales)*

The scales determine the throughput. Flow rate preselection is used to do this. *(Nominal flow rate > 0. See parameters TCON.FMAX and FLOS)*

Operating mode loading scale:

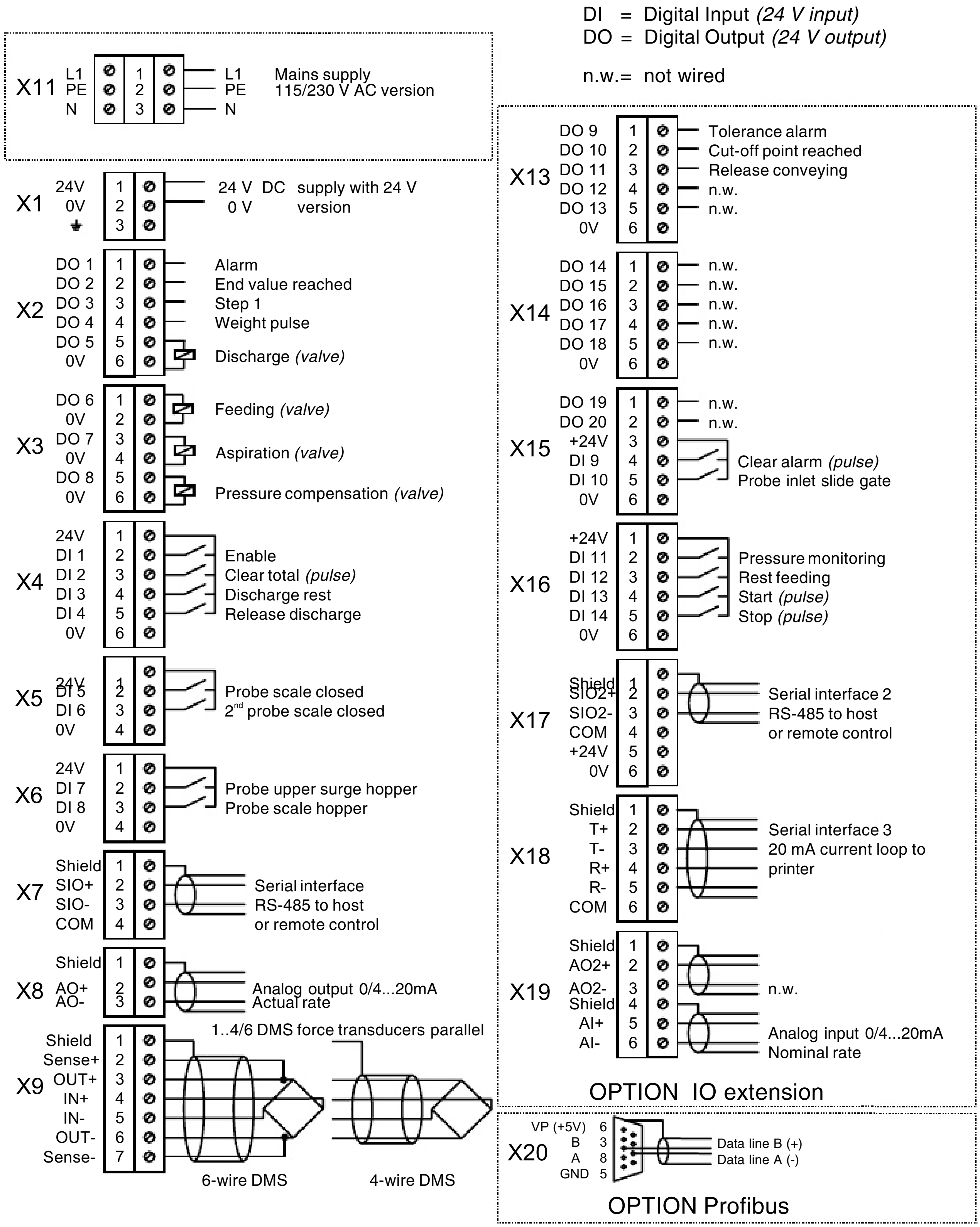
The scale feeds a preselected total weight *(via one or more dumps)*. To do this a nominal total weight is preselected *(Σ1 S)*. A nominal flow rate can also be used additionally.

4.2 Step sequence

The weight process is divided into different steps. When the scale has started, the steps 2 to 8 are processed cyclically

Step	Meaning	Description
1	Initial position	The control unit remains at this step when the scale has not been started, when no enable signal is available or when the total weight is reached.
2	Feeding	Filling the weigh hopper. The program stays at this step until the nominal dump weight *(MAIN.DWTS)* is reached or aborted with rest emptying.
3	Full weight	Storing of the full weight at standstill *(pressure compensation open).*
4	Release discharge	Wait until « Release discharge » 24 V is applied at the input.
5	Discharge	Emptying the weigh hopper. It is emptied until the weight falls below the zero tare weight *(TCON.MEWT)*
6	Close scale	Close emptying gate *(with monitor if a probe «Scale closed» is available)*
7	Empty weight	Store the empty weight at standstill. Calculation of the actual dump and total *(pressure compensation open).*
8	Batch end	If the system works with flow rate preselection then it will wait here until a new cycle is started. *(Waiting for scale – upper surge hopper probe)*
0	Calibration	The calibration switch S2 is at «CAL». It is only possible to adjust the weight and change some parameters at this step.

4.3 Connection



24 V inputs		
Terminal	**Signal**	**Description**
DI 1	Enable [1]	24 V = Enable feeding *(see also par. SYS.EIN)*
DI 2	Clear total [2]	24 V pulse min. 200 ms. The total weight (∑) is set to 0.
DI 3	Rest discharge [3]	24 V pulse min. 200 ms. Complete discharge of the scale. The signal is only accepted in step 1 and 2.
DI 4	Release discharge [4]	24 V = Release discharge *(lower surge hopper uncovered)* With unused input TCON.DIPB = OFF.
DI 5	Probe scale closed [5]	24 V = Scale closed / 0 V = scale open If no probe is available, parameter TCON.CTST = OFF.
DI 6	2nd probe scale closed [6]	24 V = Scale closed. If both probes are available, the parameter TCON.CTST = ON1+2
DI 7	Probe upper surge hopper [7]	24 V = Probe uncovered *(see TCON.IMOD)*
DI 8	Probe scale hopper [8]	24 V = Probe uncovered *(unused TCON.HLPB = OFF)* 0 V = Probe covered -> feed abort
DI 9 1)	Clear alarm [9]	24 V pulse min. 200 ms. Alarm acknowledgement *(see alarms)*
DI 10 1)	Probe inlet slide gate [23]	24 V = Inlet slide gate closed / 0 V = Open *(unused TCON.INPB = OFF)*
DI 11 1)	Pressure monitoring [24]	24 V = Compressed air pressure O.K. *(unused TCON.PRES = OFF)*
DI 12 1)	Rest feeding [13]	24 V = Carry on feeding after the nominal total weight is reached *(pulse or continuous signal)*. If nominal total weight, with cut-off point and operation mode «Feeding only if the probe is covered» are selected then the probe in the surge hopper is bridged over once the cut-off point is reached. If nominal total weight and operation mode «Feeding only if the probe is covered» are not selected then the probe surge in the hopper is bridged over. The function is reset internally with Stop or rest discharge.
DI 13 1)	Start [25]	24 V pulse min. 200 ms. Start feeding process from step 1. Only in operation if TCON.STRT=ON or STORE.
DI 14 1)	Stop [16]	24 V pulse min. 200 ms. Stops feeding process after the first discharge and then remains at step 1 Only in operation if TCON.STRT=ON or STORE.
No terminal	Remote [10]	Switching local/remote if the function is enabled according to parameter SYS.RIN. 0 V = Local *(nominal value display)* / 24 V = Remote *(nominal value external)*

1) On IO extension option
2) Wired ex works
[ ] Signal number *(for reassignment of the inputs/outputs, see SYS.DI/DO)*

24 V outputs		
Terminal	**Signal**	**Description**
DO 1	Alarm [1]	0 V = Alarm if SYS.AINV = ON
DO 2	End value reached [2]	24 V = The nominal weight (∑) was reached. The output is reset with «Clear total».
DO 3	Step 1 [3]	24 V = The control unit is in step 1
DO 4	Weight pulse [4]	24 V pulse according to SYS.WIMP
DO 5	Discharge [5]	24 V = Discharge valve activated 2)
DO 6	Feeding [6]	24 V = Valve inlet gate activated 2)
DO 7	Aspiration [7]	24 V = The aspiration gate is at aspiration 2)
DO 8	Pressure comp. [8]	0 V = Pressure compensation gate is set to pressure comp. 2)
DO 9 1)	Tolerance alarm [9]	24 V = The actual flow rate of the scale is outside the set tolerance limits MAIM.TOL+, MAIN.TOL–
DO 10 1)	Cut-off point reached [10]	24 V = The preselected cut-off point *(MAIN.CUTW)* has been reached.
DO 11 1)	Conveying release [11]	24 V = When the scale has been started and no alarm is present *(can be used for the starting of feeding elements)*

1) On IO extension option
2) Wired ex works
[ ] Signal number *(for reassignment of the inputs/outputs, see SYS.DI/DO)*

4.4 Commissioning

Basically the parameters are set at the factory in order to make the scale fully operationable.

Note:
To change most parameters the control unit must be in ***step 1*** *(current step see WT). For this purpose the scale has to be stopped. This is achieved by pressing the «F» key at the parameter WT (rest discharge) until the scale is in step 1. (Or by pressing the «F» key at the parameter FLO, if the stop function has been released at TCON.KSTO)*
In order for the scale to restart afterwards the «F» key has to be pressed at the parameter FLO.

Check or set the following points:

- The parameter SYS.TYP must be set to **DUMP** *(dump scale)*.

  (only adjustable if calibration switch is in the CAL position)
- If a lower surge hopper probe is connected at the **discharge enable input** or the signal is output by the system control unit TCON.DIPB = ON must be input on the parameter.
- If the **nominal values** for the scale are set **remotely** then the parameter SYS.REM must be set accordingly *(see chapter 3.7.5)*.
- Check whether the right nominal weight is entered for the dumping at parameter MAIN.DWTS. The value must be reduced if the scale is always overfilled in normal mode.

4.5 Parameters and alarms

4.5.1 Parameter overview

Operation parameters

FLO / FLOS Actual/nominal rate
Σ1 / Σ1S Actual/nominal total weight
CUTW Cut-off weight
Σ2 Non-erasable total weight
WT Current scale weight

Parameter groups

MAIN Main group	TCON Scale type configuration	SYS		SERV
DWT / DWTS Act./nominal dumping	MAXW Max. dump size			
LSU Last total weight	MINW Min. dump size			
FWT Full weight	MINT Small discharge amount			
EWT Empty weight	FMAX Maximum flow rate			
TOL+ Rate tolerance positive	FAVG Average actual rate			
TOL– Rate tolerance negative	MEWT Maximum zero tare			
SCNT Total counter	CTWT Corr. nom. weight target.			
DCNT Dump counter	CTST Scale closed test			
DC2 Dump count. n.-erasable	IMOD Probe upper surge hopper			
RPRO Rest feeding	HLPB Probe weigh hopper			
DTST Test stop	DIPB Probe lower surge hopper			
	INPB Inlet slide gate			
	PRES Pressure monitoring			
	FAO Flow rate analog output			
	ACUT Automatic cut-off point			
	CCYC Control cycle			
	TARD Delay auto. rest empty.			
	TDMA Max. emptying time			
	TDMI Min. emptying time			
	TINT Intermed. emptying time			
	TDAP Delay vent. after feed.			
	TDAD Delay vent. after empty.			
	TASP Aspiration time after feeding			
	TBLK Feeding block time			
	TSTA Max. standstill time			
	HTYP Host protocol type			
	STRT Start/stop inputs			
	DSTO Stop after rest discharge			
	KSTO Release «Stop» key			
	FDP Performance format OPTO22			

4.5.2 Parameter description

Code	Operation parameters		Input range
FLO / FLOS	Actual and nominal rate in t/h *(or cwt./h)* Start/stop with «F» key in the local mode *(Stop only possible if TCON.KSTO=ON)*		0…FMAX [0]
$\Sigma 1$ / $\Sigma 1S$	Actual and nominal total weight in kg Delete actual total weight with the «F» key $\Sigma 1$ is flashing once a nominal total weight is reached	1)	0…99’999’999 [0]
CUTW	Cut-off weight in kg *(only for nominal total weight > 0)*		9999
$\Sigma 2$	Non-erasable total weight in kg	1)	
WT	Current scale weight in kg with step number. Rest discharge with «F» key. Stop after rest discharge if TCON.DSTO=ON or Stop after rest discharge in the local mode if the «F» key is pressed during the entire rest discharge. At overload lines appear at the top, at underload at the bottom for invalid converter value in the middle.	1)	

1) Unit according to UNIT; decimal point according to DIV

Code	Parameter group MAIN *(main group)*		Input range
DWT / DWTS	Actual and nominal dumping in kg	1)	MINW…MAXW [30.00]
LSU	Last total weight in kg	1)	
FWT	Full weight in kg *(saved full weight in step 3)*	1)	
EWT	Empty weight in kg *(saved empty weight in step 7)*	1)	
TOL+	Rate tolerance positive in t/h. Actual TOL+ = 0, no rate monitoring takes place with respect to overrate.		0…4’000.000 [0.000]
TOL–	Rate tolerance negative in t/h. Actual TOL– = 0, no rate monitoring takes place with respect to underrate.		0…4’000.000 [0.000]
SCNT	Total counter. Is increased by 1 for every total.		
DCNT	Dumping counter erasable. Becomes 0 with clear total.		
DC2	Dumping counter non-erasable		
RPRO	Rest feeding *(switch on/off with «F» key)* The rest feeding is switched off automatically again after every rest discharge or stop.		OFF/ON [OFF]
DTST	Test stop *(checking full and empty weight)* The current scale weight is displayed. A test stop is activated by pressing the «F» key. The scale stops at full weight and at empty weight whereby the scale continues operating every time the «F» key is pressed.		

1) Unit according to UNIT; decimal point according to DIV
[ ] Standard setting

Code	Parameter group TCON *(type specific configuration)*		Input range
MAXW	Maximum dump size in kg	2) 1)	MINW…99999 [50.00]
MINW	Minimum dump size in kg	2) 1)	100…MAXW [10.00]
MINT	Smallest discharge amount in kg	2) 1)	100…99999 [20.00]
FMAX	Maximum flow rate in t/h *(or cwt./h)* 0 = Rate control switched off > 0 = With rate control. The maximum rate limits the input range of the nominal rate FLOS.		0; 0.100…4'000.000 [0.000]
FAVG	Average of the actual flow rate Number of flow rate values for the floating average of the actual flow rate FLO.		1…10 [4]
MEWT	Maximum zero tare in kg	2)	10…99999 [3.00]
CTWT	Correction of nominal weight targeting 0 = Nominal weight targeting switched off > 0 = With nominal weight targeting. Correction of the internal cut-off point according to the set value.		0.0…1.0 [0.5]
CTST	Scale closed test. Monitoring of the discharge gate with one or two limit switches «F» key: Error counter alarm 22 OPEN	1)	OFF/ON1/ON1+2 [OFF]
IMOD	Operating mode upper surge hopper probe OFF = Upper surge hopper probe *(input)* has no function FULL = Feeding from the full, only if the probe is covered FULLR = Feeding from the full, only if the probe is covered With discharge test *(24 V, «F» key, serial)* the upper surge hopper is automatically emptied as well. INT = Feeding with intermediate discharge for the time TINT if the probe is covered.		OFF…INT [OFF]
HLPB	Weigh hopper full probe *(ON = probe available)*		OFF/ON [OFF]
DIPB	Lower surge hopper empty probe OFF = Input «Release discharge» not used *(no probe or control signal connected)*. Emptying always possible. ON = Input «Release discharge» is used Emptying is enabled in the following 2 cases: – if 24 V is applied to the input and REM is not REMP – if 24 V is applied to the input and REM = REMP and the Release discharge is applied via Profibus ON_P = Release discharge via Profibus, input «Release discharge» not used. Emptying is enabled in the following 2 cases: – if REM = REMP and the Release discharge is applied via Profibus – if REM is not REMP *(e.g. in case of changeover to LOC)*		OFF/ON/ON_P [OFF]
INPB	Monitoring the inlet slide gate with a probe «F» key: Error counter alarm 32 IOPEN	1)	OFF/ON [OFF]
PRES	Compressed air monitoring with a pressure switch «F» key: Error counter alarm 33 PRES	1)	OFF/ON [OFF]

Code	Parameter group TCON *(type specific configuration)*	Input range
FAO	Flow rate analog output AO and analog input AI in t/h *(rate at 20 mA)*	0.000…4’000.000 [10.000]
ACUT	Automatic cut-off point Automatic determination of the cut-off point with intermediate stop in loading operation *(with nominal total weight pre-set)*.	OFF/ON [OFF]
CCYC	Control cycle 1) The empty weight is only measured every nth dump. If CCYC = 1, the empty weight is measured at every dump.	1…20 [1]
TARD	Delay time automatic rest discharge in sec 0 = No automatic rest discharge > 0 = If no more product is fed, automatic rest discharge is carried out after the set time	0…600 [0]
TDMA	Maximum discharge time in sec	1.0…99.9 [15.0]
TDMI	Minimum discharge time in sec	0.1…9.9 [2.0]
TINT	Intermediate discharge time in sec	0.1…9.9 [2.0]
TDAP	Delay time ventilation after feeding in sec	0.1…9.9 [1.0]
TDAD	Delay time ventilation after emptying in sec	0.1…9.9 [0.5]
TASP	Aspiration time after feeding in sec	0.1…9.9 [0.6]
TBLK	Feed blocking time in sec	0.1…9.9 [1.5]
TSTA	Maximum standstill time in sec	0.1…9.9 [5.0]
HTYP	Host protocol type STD = Standard host protocol *(like MWEE)* MWET = Host protocol MWET *(e.g. total 9 digits)* ETPC = Interface to the MWET-PC *(from version 27A 8 bit data)* ETPCNP = Interface to the MWET-PC without parity-bit	STD/MWET/ETPC [STD]
STRT	Start/stop inputs *(pulse)* OFF = Inputs have no function ON = The scale is started and stopped with the inputs. *(Function like MWES/MWET)* STORE = Inputs are operating. The Start signal is saved, also if release = 0 V	OFF/ON/STORE [OFF]
DSTO	Automatic stop after rest discharge or if final value is reached	OFF/ON [OFF]
KSTO	Released Stop key *(«F» key parameter FLO in the local mode)*	OFF/ON [OFF]
FDP	Performance format at the serial interface *(OPTO22)* OFF = 1 kg/h resp. 1 lb/h ON = 10 kg/h resp. 0.1 cwt/h	OFF/ON [OFF]

1) Only adjustable or executable if the calibration switch is in the «CAL» position
2) Unit according to UNIT; decimal places according to DIV
[ ] Standard setting

4.5.3 Alarms

Alarm number	Meaning	Remedy
1…18	**General alarms**	See chapter 3.8 General alarms
20 MOVE	**No standstill** The scale weight is not steady	– Strong vibrations of the scale 4) – Inlet or outlet gate not tight – Air disturbances *(check function of air flaps)* – Set max. standstill time or standstill tolerance higher *(TCON.TSTA or ADC.MOT)*
21 DTIME	**Emptying time** The scale is not emptied within the max. emptying time	– The discharge gate does not open or only partly 4) – Set the max. emptying time higher *(TCON.TDMA)*
22 OPEN	**Scale open** The outlet gate is open at feeding start, during feeding or after discharging	– The discharge gate is jammed 4) – The «Scale closed» probe is defective – The discharge gate is forced open by the product – TCON.CTST set incorrectly
23 HOPP	**Probe scale hopper** The probe is still covered after emptying	– Product sticking to the probe 4) – Probe defective – TCON.HLPB set incorrectly
26 TOL	**Tolerance** The actual rate is outside the preselected tolerance *(not at alarm output)*	– Find cause of the wrong flow rate 4) – Increase tolerances *(MAIN.TOL+/TOL–)*
30 ZERO	**Zero tare** At the start the current scale weight is higher than the zero tare *(TCON.MEWT)*	– Inlet slide gate is leaking 4) – Scale has been incorrectly adjusted If there is product in the weigh hopper the alarm must be acknowledged with rest discharge.
32 IOPEN	**Inlet slide gate** Open at initial position, during weight acquisition or discharging Closed during feeding	– Inlet slide gate is jammed 4) – Probe faulty – TCON.INPB set incorrectly In the step feeding *(2)* the alarm can only be acknowledged with rest discharge.
33 PRES	**Air pressure drop**	– Check compressed air 4) – TCON.PRES set incorrectly

4) Acknowledgement depending on parameter SYS. ACLR

ACLR = OFF: Automatic acknowledgement when the cause of the alarm is eliminated or when changing to step 1

ACLR = ON: No automatic acknowledgement. Acknowledgement only with key or 24 V input «Clear alarm»

5 Differential dosing scale *(DIFF, DIFFG, DIFFM)*

5.1 Description

Operating mode flow control *(TCON.MODE = FLOCON)*

Control of discharge flow rate *(actual flow rate)* to the set nominal flow rate *(FLOS)*. The actual flow rate is calculated by measuring the weight loss in the weigh hopper. If the weight in the weight hopper reaches a minimum weight *(lower switching point TCON.LLEV)*, an output is activated for the automatic product feed. Discharging continues during refilling. This operating mode is possible with or without nominal weight. At nominal weight = 0 feeding is endless.

Operating mode flow meter *(TCON.MODE = FLOMET)*

Measuring of a given product flow *(weight and flow rate)*. The discharge flow rate is controlled according to the feed flow rate so that there is a constant level in the weight hopper *(TCON.NLEV)*.

Operating mode small batch feeding *(TCON.MODE = CHARGE)*

Accurate weight feeding to a preselected nominal weight *(Σ1 S)*. The screw speed remains constant according to the given nominal flow rate. If the nominal weight cannot be fed with a hopper filling, refilling takes place automatically. Discharging does not continue during refilling. This operating mode is therefore very accurate because the weight is measured before and after dosing at standstill of the screw.

Operating mode small batch feeding or flow controller *(TCON.MODE = CHAFLO)*

According to the nominal weight it is selected automatically whether the device operates in the «Small batch feed» or «Flow control».

Small nominal weights contained in a hopper filling are fed according to the «Flow control» principle.

In this mode the nominal weight is always fed free of interruption.

Admixture mode *(TCON.ADDM = ON)*

The admixture mode is not an autonomous operating mode and can only be selected additionally in the «Flow control» mode. In the admixture mode a product is fed as a percentage of a master scale. Both nominal values for the flow rate of the master scale and the admixture percentage are calculated for the internal nominal flow rate. The flow rate of the master scale is given through the analog current interface on the I/O extension option.

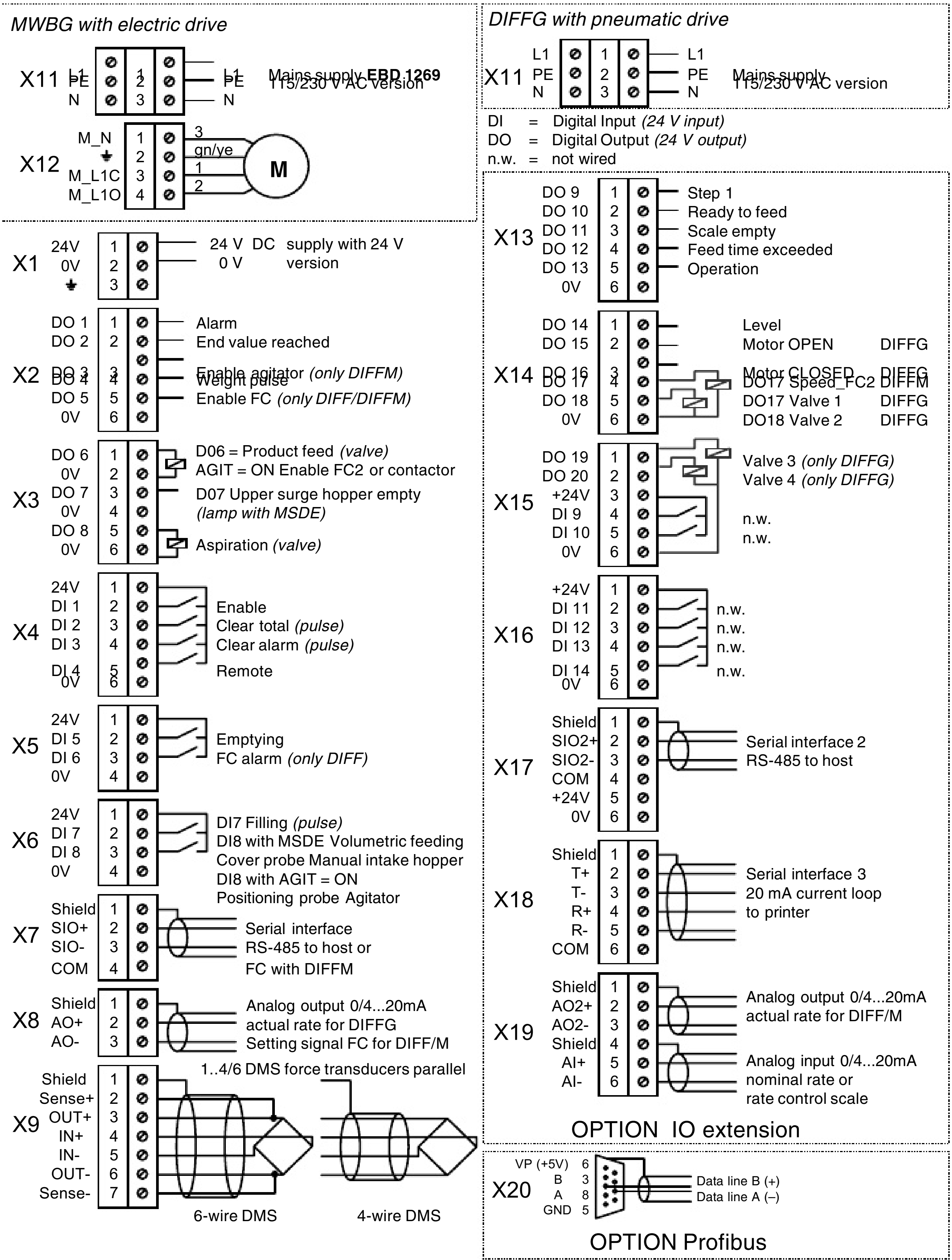
5.2 Stepping sequence

The weighing process is divided into different steps. In the operating modes «Flow controller» and «Flowmeter», the steps 2 to 6 are processed cyclically.

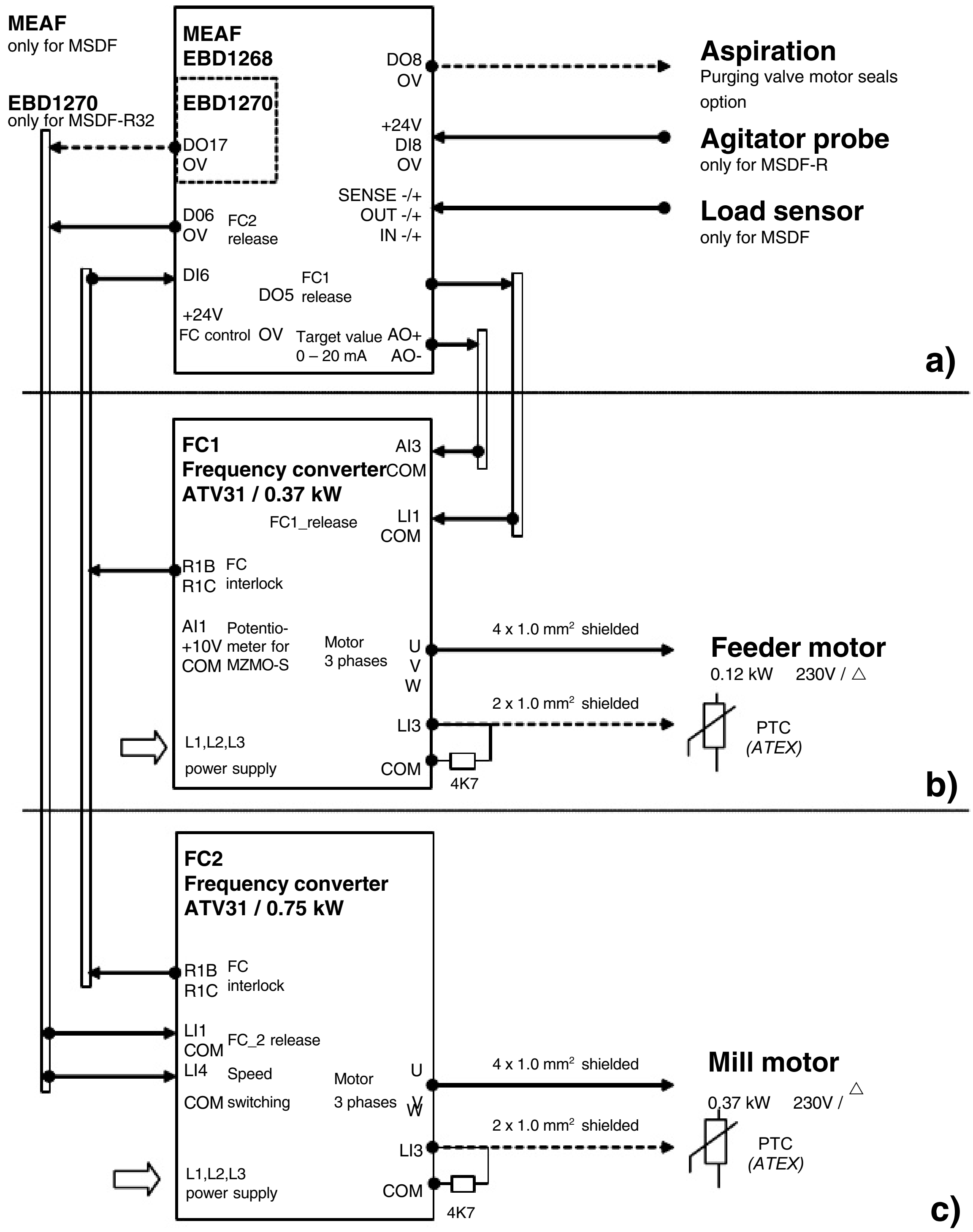
Step	Meaning	Description
1	Initial position	Initial position for all operating modes. The control unit stays at this step if there is no feed enable, no filling or emptying requested or the total weight is reached.
2	Refill hopper	By setting the output «Product feed», product is requested for filling the scale up to the upper switching point. A time monitor is installed for refilling. Discharging continues during refilling.
3	Venting lag time	The «Aspiration» output is activated according to the set «Venting lag time». The step is only executed if the weight hopper was refilled in advance in step 2.
4	Stabilizing time	Waiting time according to the selected «Stabilizing time» until measuring starts.
5	Measuring time	During the measuring time the measuring points are saved for calculation of the flow rate. The measuring time and the number of measuring points are selected automatically according to the required nominal flow rate and the actual control range.
6	Calculations	Calculation of the actual flow rate and the total weights. In the «Flow controller» mode the setting signal for the screw speed is calculated and output according to the nominal/actual deviation.
7	Emptying	Emptying takes place at a constant flow rate of the weight hopper.
8	Filling	Automatic filling of the scale by setting the output «Product feed». In the DIFF and DIFFM types the screw is also filled if a nominal flow rate is preselected.
9	Feed small batch	Accurate weight feeding to a preselected nominal weight. The screw speed remains constant according to the specified nominal flow rate.
10	Leak test	Ex works setting at Bühler *(only DIFFG)*
11	Valve test	Functional test of the valve assembly *(activation of the metering slide gate)*. This test consists of a visual check of measuring distances *(basic position to metering slide gate positions according to test step; only DIFFG)*

Step	Meaning	Description
12	Agitator test	DIFFM and TCON.AGIT = ON **ATIP: Agitator functions** Rotation direction test – agitator motor Step operation for removal of the agitator blade when cleaning Setting the positioning probe => move into nominal position. Start by pressing the F-key => agitator blade moves out of the nominal position. **Rotation direction test** When the agitator is moving, MOVE is shown on the display. Clockwise turning of the agitator blade when viewed from above **Step operation** Press the + key briefly The agitator blade moves a short distance. If the probe remains uncovered, XPOS stands for any position of the agitator blade on the display. **Move to nominal position** Press and hold the + key down until the position probe is covered. SPOS is now shown on the display for nominal position reached. *Note:* When the nominal position has been reached, the function test ATIP remains interrupted even if the + key is pressed again. This function is not reactivated until the F-key is pressed to exit the program.
0	Calibrate	The calibration switch is in the «CAL» position. It is only possible to calibrate the scale and change some parameters at this step.

5.3 Connection



5.4 Wiring designs



Scales		Feeders		Design type	Installation / attachment
MSDF_R	a + b + c	MZMO_R	b + c	R = Refill	Built-in housing
MSDF_S	a + b	MZMO_S	b	S = Standard	Carrier plate

Only for scale type MSDF-R → MEAF : TCON.AGIT = ON *(with agitator probe)*.

If no thermistor is present in the motor, ***ATV31*** ► Set **Flt** menu parameter / EPL = **nO** or jumper LI_3 to +24 V.

5.5 Control system connection

Mains connection ►	*ATV31 3-phase mains*	*MEAF 1-phase mains*	*Scale*
MSDF- R *32* FC1 + FC2 IP20 Diagram ESP-84231 UL ESP-84236	**X101** 4 5 6 7 **L1 L2 L3 PE** 3 x 200 – 240 V 50/60 Hz 3 x 380 – 500 V 50/60 Hz	**X101** 8 9 10 **L N PE** 1 x 115 V 50/60 Hz 1 x 230 V 50/60 Hz	**Scale**
MSDF- R *20* FC1 + FC2 IP20 Diagram ESP-84232 UL ESP-84237	**X101** 4 5 6 7 **L1 L2 L3 PE** 3 x 200 – 240 V 50/60 Hz 3 x 380 – 500 V 50/60 Hz	**X101** 8 9 10 **L N PE** 1 x 115 V 50/60 Hz 1 x 230 V 50/60 Hz	
MSDF- S *20 + 32* FC IP54 Diagram ESP-84234	Frequency converter ATV31 **Switch S1** **T1 T2 T3 PE** L1 L2 L3 PE_ATV31 3 x 380 – 500 V 50/60 Hz	**X201** Frequency converter 1 2 3 **L N PE** 1 x 115V 50/60 Hz 1 x 230V 50/60 Hz	
Mains connection ►	*ATV31 1-phase*	*MEAF 1-phase mains*	
MSDF- S *20 + 32* FC IP54 Diagram ESP-84235	Frequency converter **Switch S1** **T1 T2 PE** L1 N PE_ATV31 1 x 200 – 240 V 50/60 Hz	Frequency converter ATV31 **Switch S1** **L1 L2 PE** 1 x 200 – 240 V 50/60 Hz	

Mains connection ►	*ATV31 3-phase mains*	*ATV31 Release +24 V* Frequency converter ATV31	*Feeder*
MZMO- R *20 + 32* FC IP54 Schema ESP-84233 UL ESP-84238	**X101** 4 5 6 7 **L1 L2 L3 PE** 3 x 200 – 240 V 50/60 Hz 3 x 380 – 500 V 50/60 Hz	**X103** Signal or Bridge external or local **COM LI1 +24 V** 3 2 1	**Dosing system**
MZMO- S *20 + 32* FC IP54 ESP-84241	Frequency converter ATV31 **Switch S1** **T1 T2 T3 PE** L1 L2 L3 PE_ATV31 3 x 380 – 500 V 50/60 Hz	Signal or Bridge external or local **COM LI1 +24 V**	
Mains connection ►	*ATV31 1-phase*	*ATV31 Release +24 V*	
MZMO- S *20 + 32* FC IP54 ESP-84240	Frequency converter **Switch S1** **T1 T2 PE** L1 N PE_ATV31 1 x 200 – 240 V 50/60 Hz	Signal or Bridge external or local **COM LI1 +24 V**	

Note:
The factory supplies the corresponding scheme with each control unit.
PoD B-Net-Link to all original schemes under: MSDF-MZMO No. 66677

24 V inputs		
Terminal	**Signal**	**Description**
DI 1	Enable [1]	24 V = Feed enabled *(see also par. SYS.EIN)*
DI 2	Clear total [2]	24 V impulse min. 200 ms. The weight total (∑) is set to 0 *(only if the control, unit is in step 1)*
DI 3	Clear alarm [9]	24 V impulse min. 200 ms. Alarm acknowledgement *(see alarms)*
DI 4	Remote [10]	Switching Local/Remote if the function is enabled according to parameter SYS.RIN. 0 V = Local *(nominal values display)* 24 V = Remote *(nominal value external)*
DI 5	Emptying [3]	24 V = Emptying the scale as long as the signal is applied. Emptying is only possible from step 1.
DI 6	FC alarm [11]	24 V = Fault feeding drive if TCON.FIIN = OFF 2) 0 V = Fault feeding drive if TCON.FIIN = ON *(only with DIFF, DIFFM)*
DI 7	Filling [15]	24 V impulse min. 200 ms. The scale is filled with product.
DI 8	Volumetric feeding [13]	24 V = Volumetric feeding with a constant discharge rate only with DIFF, DIFFM and operation mode flow meter. The signal should be present, e.g. during a manual refill *(directly into the weigh hopper).*
	Positioning probe	24 V = Probe for agitator positioning covered ≙ in position Agitator blade must cover the product opening completely DIFFM : MSDF-R *(TCON.AGIT = ON)*

24 V outputs		
Terminal	**Signal**	**Description**
DO 1	Alarm [1]	0 V = Alarm if SYS.AINV = ON
DO 2	End value reached [2]	24 V = The nominal weight (∑) has been reached. The output is reset with «Clear total».
DO 3	Enable agitator [19]	24 V = The agitator is activated according to parameter TCON.AOFF *(only in DIFFM)*
DO 4	Weight pulse [4]	24 V pulses according to SYS.WIMP and SYS.IMPM
DO 5	Enable FC [5]	24 V = Feed motor enabled *(only DIFF and DIFFM)*
DO 6	Product feed [6]	24 V = Open inlet gate 2) DIFFM : MSDF D32-R TCON.AGIT = ON enable FC2 DIFFM : MSDF D20-R TCON.AGIT = ON enable contactor
DO 7	Upper surge hopper empty [14]	24 V = The scale could not be filled in the max. filling time *(TCON.TFIL)* In the case of the flow meter the output is set if the product supply has stopped.
DO 8	Aspiration [7]	24 V = The aspiration/airlock is at Asp. 2)
DO 9 1)	Step 1 [3]	24 V = The control unit is in step 1
DO 10 1)	Ready to feed [15]	24 V = The scale is full
DO 11 1)	Scale empty [16]	24 V = The weigh hopper is empty *(Alarm 24 EMPTY)*
DO 12 1)	Feed time exceeded [17]	24 V = The maximum feed time was exceeded during the feeding of small batches or emptying
DO 13 1)	Operation [18]	24 V = The actual rate is greater than 0
DO 14 1)	Level [20]	24 V = Current scale weight higher than level weight *(TCON.NLEV)*
DO 15 1)	Close [25]	24 V = Close motoric metering slide valve *(only MWBG)*
DO 16 1)	Open [26]	24 V = Open motoric metering slide valve *(only MWBG)*
DO 17	Valve 1 FC2 [21]	24 V = Air supply *(de-energized closed)* 3) 4) 24 V = MSDF D32-R Agitator with FC2 *(DIFFM : TCON.AGIT = ON)*
DO 18	Valve 2 [22]	24 V = Air supply, throttled *(de-energized closed)* 3) 4)
DO 19	Valve 3 [23]	24 V = Air bleed, throttled *(de-energized open)* 3) 4)
DO 20	Valve 4 [24]	24 V = Air bleed *(de-energized open)* 3) 4)

1) On IO extension option
2) Wired ex works
3) In the case of pneumatic metering slide valve wired ex works
4) Only with DIFFG and TCON.RVA = ON
[ ] Signal number *(for reassignment of the inputs/outputs, see SYS.DI/DO)*

5.6 Commissioning

Basically the parameters are set at the factory so that the scale is functional.

Check or set the following points:

- The parameter SYS.TYP must be set to **DIFFG, DIFF or DIFFM** according to the differential feed scale type.
  (Only adjustable if the calibration switch is in the CAL position)
- It must be checked whether the scale is set to the **correct operation mode** at parameter TCON.MODE *(flow controller, flow meter, small batch feed)*. See also chapter «Description».
- If the scale is used as an **admixture scale**, the parameter TCON.ADDM must be set to ON and the rate of the control scale set at parameter TCON.MAST.
- If the **nominal values** for the scale are set **remotely** then the parameter SYS.REM must be set accordingly *(see chapter 3.7.5)*.
- If only **one product** is being processed, the specific weight must be entered at parameter REC.DENS *(REC.NR = 0)*.

Different products

If products of different specific weight are fed, recipes should be used. One recipe memory should be used for every product.

The scale should be run in with every product.
Procedure for every product *(only at DIFF and DIFFM)*:

- Input of the specific weight at parameter REC.DENS for recipe number 0.
- Start feeding with as high a rate as possible *(50%…100 % of the maximum rate)*.
- The control behaviour can be observed at parameter MAIN.MON. If the control for several measuring cycles is in the fine tolerance *(display «F»)*, the setting can be saved in a recipe. This is done by entering the desired recipe number at parameter MAIN.SLAVE.
- The device should be stopped and restarted for checking. Now the rate nominal value should be reached as quickly as possible.

5.7 Parameters and alarms

5.7.1 Parameter overview

Operation parameters

FLO / FLOS	Actual/nominal rate
Σ1 / Σ1S	Actual/nominal total weight
Σ2	Non-erasable total weight
WT	Current scale weight
ADD / ADDS	Actual/nominal admix. process

Parameter groups

MAIN – Main group

LOAD	Load recipe
SAVE	Save recipe
MON	Control info
CTRL	Control
FILL	Fill
ATIP	Agitator test
LK	Leak test
VA	Valve test

REC – Recipes

NR	Recipe number
DENS	Spec. weight
FACT	Gain factor
CFAC	Correction gain factor
CWT	Cut-off small batch
PROT	Write protect. recipe

TCON – Scale type config.

MODE	Operating mode
ADDM	Admixture mode
EMOD	Empty at nom. weight
AGIT	Agitator Position.probe
RVA	Valves or motor
VOL	Vol. scale hopper
HLEV	Upper switching point
LLEV	Lower switching point
NLEV	Nominal level flow meter
FMAX	Maximum rate
FMIN	Minimum rate
MAST	Max. rate control scale
CTOL	Coarse tolerance
FTOL	Fine tolerance
RCOR	Control correction
TPOS	Positioning time
TFIL	Max. filling time
TASP	Aspiration time
TSTA	Stabilization time
SMES	Sensitivity MEAS
TMES	Min. max. meas. time
TFGO	Gate setting time until product flows
TFG	Max. gate setting time
FTYP	Type rate
FCOR	Rate correction
AOFF	Cutout mode agitator
MIX	Mixer frequency
FIIN	Input fault drive
KSTO	Release Stop key

SYS

5.7.2 Parameter description

Code	Operation parameters	Input range
FLO / FLOS	Actual and nominal flow rate *(in t/h or cwt./h at DIFF and DIFFG / in kg/h or lb./h at DIFFM)* Start/stop with «F» key in local mode *(With operation mode FLOMET stop only possible if TCON.KSTO=ON)*	0…999.999
Σ1 / Σ1S	Actual and nominal total weight in kg 1) Clear total with «F» key in local mode Σ1 is flashing once a nominal total weight is reached	0…99’999’999
Σ2	Non-erasable total weight in kg 1)	
WT	Current scale weight in kg with step number 1) Discharging start/stop by pressing the «F» key. *(step 7)* At overload lines appear at the top, at underload at the bottom, at invalid converter value in the middle.	
ADD / ADDS	Actual and nominal admixture percentage in % *(only in operating mode TCON.ADDM = ON)*	0…99.9999

1) Unit according to UNIT; decimal point according to DIV

Code	Parameter group MAIN *(main group)*	Input range
LOAD	Load recipe with «F» key or by changing the number *(from the memory to working recipe 0)*	1…50
SAVE	Save recipe with «F» key or by changing the number *(save working recipe 0 in the memory)* If «NO-SAVE» appears another number has to be selected or the write protection in the recipe has to be cancelled first *(REC.PROT=OFF)*	1…50
MON	Control info Control range *(letter left)* F = within fine tolerance C = between fine and coarse tolerance O = outside coarse tolerance Control deviation actual flow rate-nominal flow *rate (number on right)*	
CTRL	Control *(only at DIFF, DIFFM and MODE = FLOCON)* OFF = Volumetric feeding according to nominal flow rate without control of the actual flow rate ON = Control of the actual flow rate	OFF/ON [ON]
FILL	Start filling by pressing the «F» key *(step 8)* The current scale weight with step is displayed.	
ATIP	Activation of the agitator: for step operation or moving into nominal position *(only DIFFM and TCON.AGIT = ON).* Start with «F» key.	MOVE XPOS / SPOS
LK	Ex works setting at Bühler *(only DIFFG)*	–
VA	Functional test of the valve assembly *(activation of the metering slide gate; only DIFFG)* Start / stop with «F» key in local mode	FO ... WAIT SO ... WAIT SC ... WAIT ... FC

[ ] Standard setting

Code	Parameter group REC *(recipes)*	Input range
NR	Recipe number This recipe number is only for viewing or changing recipes. 0 = working recipe.	0…50 [0]
DENS	Specific weight in kg/l *(bulk density)* 2)	0.03…3.00 [0.75]
FACT	Gain factor 1) 2) A correctly set gain factor ensures that the right feeding speed is selected immediately. Automatic setting if CFAC = ON *(The automatic setting is only carried out when the operation takes place within the fine tolerance and the capacity is ≥25% of FMAX and the product quantity > LLEV.)*	10.0…200.0 [100.0]
CFAC	Automatic correction of the gain factor 1) 2)	OFF/ON [ON]
CWT	Cut-off weight in small batch feeding 1) 2)	0…9999 [0]
PROT	Write protection recipe with MAIN.SAVE 2)	OFF/ON [OFF]

1) Only with DIFF and DIFFM
2) The recipe number is displayed between code and value
[ ] Standard setting

Code	Parameter group TCON *(type specific configuration)*	Input range
MODE	Operating mode FLOCON = Flow controller FLOMET = Flow meter CHARGE = Small batch feeding 1) CHAFLO = Small batch feeding or flow controller according to nominal total weight 1)	FLOCON… CHAFLO [FLOCON]
ADDM	Admixture mode	OFF/ON [OFF]
EMOD	Emptying at nominal weight When feeding to a nominal weight only as much product is refilled automatically so that the weight hopper is empty at the end.	OFF/ON [OFF]
AGIT	Agitator with positioning probe: MSDF-R *(DIFFM and TCON.AGIT = ON)*	OFF/ON [OFF]
RVA	Type of drive of the metering slide gate *(only with DIFFG)* ON = pneumatic diaphragm drive OFF = electric with motor	OFF/ON [ON]
VOL	Volume weight hopper in l *(liters)* 2) 3)	1…99999 [48]
HLEV	Upper switching point in % of VOL 2)	30…99 [95]
LLEV	Lower switching point in % of VOL 2) *(DIFFG 35 % / DIFF 55 % / DIFFM 20 %)*	1…70 [35]
NLEV	Nominal level flow meter in % of VOL 2) *(DIFFG 70 % / DIFF 80 % / DIFFM 70 %)*	30…90 [70]

1) Only with DIFF and DIFFM
2) The corresponding weight is displayed when the «F» key is pressed
3) Standard values acc. to the type taken from the table on the next page

[ ] Standard setting

Code	Parameter group TCON *(type specific configuration)* *(continued)*	Input range
FMAX	Maximum flow rate DIFF, DIFFG in t/h or cwt./h Input range of FLOS DIFFM in kg/h or lb./h 3) The set value corresponds also to the 20 mA value of the analog current interfaces. DIFFG [12000]	0.100…999.999 1.00…9999.99
FMIN	Minimum flow rate *(typical 5 % or FMAX)* DIFF, DIFFG in t/h or cwt./h Input range of FLOS DIFFM in kg/h or lb./h	0.010…99.999 0.10…999.99
MAST	Maximum flow rate control scale in t/h *(in admixture mode)*	0.1…999.999
CTOL	Coarse tolerance for flow control in % of nominal flow rate	3.0…10.0 [5.0]
FTOL	Fine tolerance for flow control in % of nominal flow rate	0.3…3.0 [1.0]
RCOR	Control correction Correction of flow rate deviations in flow controller or level control in flow meter. DIFFG [0.7]	0.4…1.0 [1.0]
TPOS	Positioning time [s] of the agitator blade. This parameter automatically adapts itself to changes in the agitator speed. The adaptation of the positioning time is only activated when the agitator is switched on for a min. of 10 s. Note: Set TFIL for a min. of 15 s	0.0…2.0 [0.5]
TFIL	Maximum refill time in s *(in operating mode FLOMET 3 s)*	3.0…99.9 [25.0]
TASP	Aspiration time after refilling in s *(in operating mode FLOMET 1.5 s/in DIFFM 3 s, DIFF 0.8 s, DIFFG 1.5 s)*	0.1…9.9 [0.3]
TSTA	Stabilization time before measurement in s *(in operating mode FLOMET 1 s / in DIFFM 3 s, DIFF 1.4 s, DIFFG 1 s)*	0.1…9.9 [1.2]
SMES	Sensitivity of the measuring fault 1.0 = insensitive 10.0 = very sensitive 0.0 = measuring fault switched off	0.0…10.0 [8.0]
TMES	Minimal *(left)* and maximal *(right)* measuring time in s These are limiting values only. The actual measuring time is calculated according to the nominal flow rate.	1…20 1…180 [2] [60]
TFGO	Gate setting time at start until product flows in ms (*only for DIFFG)*	0.1…5.0 [1.0]
TFG	Gate setting time for type rate in s *(only in DIFFG)* DIFFG [4]	3…20 [10]
FTYP	Volumetric type rate in l/h 3) Maximum flow rate for the scale size MSDG [20000]	5…999999 [30000]
FCOR	Flow rate correction factor *(100.0 = no correction)*	95.0…105.0 [100.0]
AOFF	Cutout mode agitator in % *(only in DIFFM)* The agitator is switched off for the preselected percentage *(0 = no switch off; set to 80 with 20 mm screws)*	0…90 [0]

Code	Parameter group TCON *(type specific configuration)* *(continued)*	Input range
MIX	Mixer frequency in Hz *(only with DIFFM and SYS.SIO = FIN3)*	10.00..70.00 [50.00]
FIIN	Input fault feeding drive OFF = Alarm with 24 V at the input ON = Alarm with 0 V at the input *(with DIFFM)*	OFF/ON [OFF]
KSTO	Stop key released with operation mode FLOMET *(«F» key parameter FLO)*	OFF/ON [ON]

1) Only with DIFF and DIFFM
2) The corresponding weight is displayed when the «F» key is pressed
3) Standard values acc. to the type taken from the table on the next page
[ ] Standard setting

5.7.3 Type-dependent standard values for the parameters VOL, FMAX, *(FMIN)* and FTYP

DIFFG	**MSDG-30**	**MSDG-60**	**MSDG-140**	**MSDG-200**
VOL [l]	30	60	140	200
FMAX [t/h]	12 (6)	24 (13)	60 (40)	80 (50)
FTYP [l/h]	20‘000	35‘000	85‘000	120‘000

DIFFG	**MWBG-10**	**MWBG-17**	**MWBG-48**	**MWBG-110**	**MWBG-130**	**MWBG-150**
VOL [l]	10	17	48	110	130	150
FMAX [t/h]	4 (2)	8 (4)	20 (10)	48 (30)	64 (40)	80 (50)
FTYP [l/h]	12‘000	12‘000	30‘000	64‘000	64‘000	80‘000

DIFF	**MSDA-100/100**	**MWBO-140/160**	**MWBO-280/200**	**MWBO-350/250**
VOL [l]	130	200	380	500
FMAX [t/h]	3 (2)	15 (10)	30 (30)	45 (45)
FTYP [l/h]	6‘000	30‘000	60‘000	100‘000

DIFFM **MSDE 70 / ..**											
Screw diameter [mm]	20	20	20	20	20	32	32	32	50	50	50
Gear ratio i	135	65	38	25	15	38	25	15	38	25	15
VOL [l]	60	60	60	60	60	60	60	60	60	60	60
FMAX [kg/h]	8	16	30	50	80	150	250	350	500	800	1200
FTYP [l/h] *(with 100 Hz)*	13	27	45	70	110	210	320	510	740	1130	1800

DIFFM **MSDF_R/S**										
Screw diameter [mm]	20	20	20	20	20	32	32	32	32	32
Belt set ratio	18:54	18:36	36:36	36:18	54:18	18:54	18:36	36:36	36:18	54:18
VOL [l]	9/25	9/25	9/25	9/25	9/25	9/25	9/25	9/25	9/25	9/25
FMAX [kg/h]	8	12	25	50	70	60	100	200	450	600
FMIN [kg/h]	0.75	0.75	1.5	2.0	3.0	3.0	5.0	10	20	30
FTYP [l/h] *(with 100 Hz)*	12	18	36	72	108	100	150	300	600	900

( ) The values in round brackets are for operating mode flowmeter FLOMET

5.7.4 Alarms

Alarm number	Meaning	Remedy
1…18	**General alarms**	See chapter 3.8 General alarms
23 HOPP	**Positioning probe Agitator** can no longer be positioned	– Probe faulty *(replace),* Alarm acknowledgement only with key – mechanical position set incorrectly
24 EMPTY	**Weight hopper empty** Feed enable and nominal flow rate are applied. The actual flow rate is 0 and the hopper filling is below the lower flow switching point	– Ensure product feed 4)
25 FINV	**Frequency converter** The frequency converter reports a fault *(with DIFFM one or several frequency converters)*	DIFF: – Check frequency converter 4) DIFFM: MSDE – Check display frequency converter 3) If the alarm «CAL‘ or ‚CE‘ is displayed at the frequency converter and if the fault does not disappear despite «Clear alarm», then the RS-485 interface to the converter is faulty or interrupted. – Check parameter SYS.SIO = FIN2 or FIN3 – Check parameter of converter according to MSDE manual – DIFFM: MSDF – Check parameter of converter according to MSDF manual – Check back-up fuse or wiring
26 TOL	**Tolerance** The control operates continuously *(min. 5 measuring cycles)* outside the coarse tolerance *(can also occur in very inhomogeneous product)*	– Change to «Volumetric feeding» mode, for very inhomogeneous product. 4) – The gain factor is set much too high. See commissioning with product. – Increase the coarse tolerance.
27 MEAS	**Measuring fault** During the measuring time big sudden weight fluctuations constantly occur	– Clear external influence on weight hopper *(knocks, air)*. 4)
28 MOTOR	**Feed drive** Feed enable and nominal flow rate are applied. The actual rate is 0 and the hopper filling is over the lower switching point.	– Product forms a bridge 4) – Check cabling MEAF to the feed drive – For DIFFG check pneumatic drive and piping, possibly carry out LK test and or VA test – For MWBG check or replace motors – Change basic print
29 PTIME	**Feeding time exceeded** The max. feeding time was exceeded during the small-batch feeding or during the emptying *(not on alarm output)*	– No or too low discharge rate – Incorrect specific gravity or incorrect amplification factor has been set. The alarm is automatically acknowledged with the next start or cleared with Alarm.

Alarm number	Meaning	Remedy
34 NOPRO	**Surge hopper empty *(no product supply)*** During the max. refilling time the scale could not be filled *(not on alarm output)*	– Feeding elements are empty – Increase max. refilling time *(TCON.TFIL)* The alarm is automatically acknowledged with the next start or when the scale is refilled or is cleared with Total. The alarm is also acknowledged with Stop in the operation mode FLOCON.

3) Acknowledgement with key or 24 V input «Clear alarm»
4) Acknowledgement dependent upon parameter SYS.ACLR

ACLR = OFF: Automatic acknowledgement if the cause of the alarm is eliminated or if you change to step 1

ACLR = ON: No automatic acknowledgement. Acknowledgement only with key or 24 V input «Clear alarm»

6 Bagging scale *(BAG)*

6.1 Description

The control unit feeds every bag according to a fixed process. If operation takes place with control cycle n, only every nth bag is checked after reaching optimum actual bag weights.

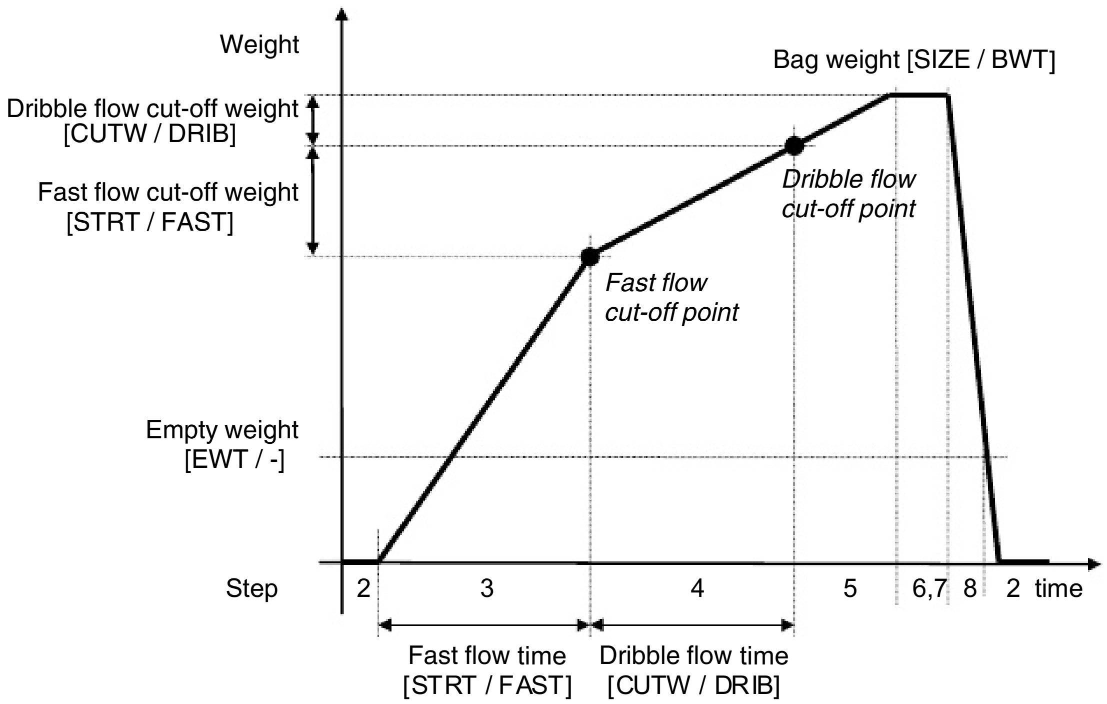
Principally it is possible to work **with or without bag preselection.** If a number of bags is entered in the nominal bag counter then this number will be fed and the output «Release bagging» switches to 0 V. With this signal a further start can be prevented externally. If 0 is entered in the nominal bag counter *(without bag preselection)* the feeding will take place endlessly.

6.2 Stepping sequence

When the scale is started, the steps 2 to 9 are processed cyclically.

Step	Meaning	Description
1	Initial position	The control unit stays in this position after an abort or in the event of an alarm.
2	Zeroing with zero check	The scale is tared to 0. The zeroing is omitted if no control cycle is executed or if the input «Continue feeding» is set.
3	Fast flow	Filling the scale with fast flow. Fast flow is ended on reaching the fast flow switch off point.
4	Dribble flow	Filling the scale with dribble flow. The dribble flow is ended on reaching the dribble flow switch-off point.
5	Nominal weight check	If a control cycle is executed, the actual bag weight is saved at standstill. The feeding values are optimized.
6	Tipping	If the weight is too small and adding is enabled *(TCON.TIP = ON)*, the missing amount is fed.
7	Release discharge	Wait until «Release discharge» 24 V is applied at the input.
8	Discharge	Empty the weight hopper. It is emptied until the weight falls below the zero tare weight *(REC.EWT)*.
9	Next start	New start when the start signal is applied.
0	Calibrate	The calibration switch S2 is at the «CAL» position. It is only possible to adjust the weight and change some parameters at this step.

Weight curve bagging scale *(control cycle)*



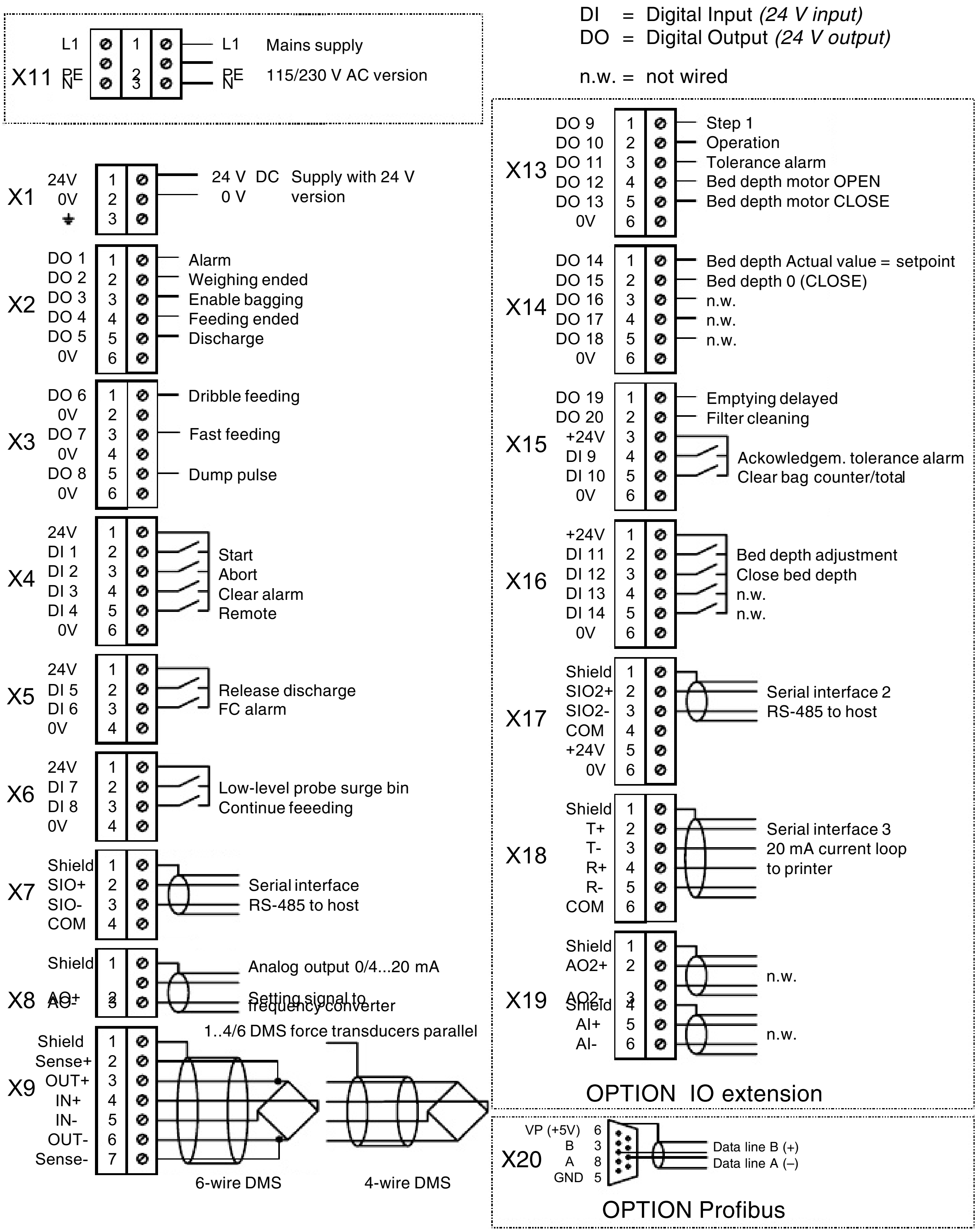
[ ] = Parameters on the display [nominal value / actual value]

The **fast flow** takes place according to weight or time *(according to parameter REC.DOSW)*. Normally operation takes place with weight feeding. Time feeding is chosen if the fast flow time is under 2 seconds. This is usually the case if no frequency converter is available and small weights *(below 20 kg)* are fed. If a frequency converter is available, the fast flow feed rate can be reduced if necessary at the REC.FFLO parameter. The fast flow time or weight value is input at the REC.STRT parameter. If 0 is entered, no fast flow is executed *(the whole bag is fed in dribble flow)*. The fast flow cut-off weight or the fast flow time REC.STRT is controlled automatically so that the dribble feeding takes place in the dribble flow nominal time REC.TDRI.

The **dribble feeding** always takes place according to weight with the dribble flow cut-off weight REC.CUTW. The dribble flow cut-off weight is controlled automatically so that accurate bag weights are produced.

Control cycles are used to increase performance. This means that only every nth bag is checked as soon as the bag weight is within the desired tolerance. In unchecked bags, steps 2, 5 and 6 *(zeroing, nominal weight check and adding)* are dropped. *(Normal setting REC.CCYC = 10)*

6.3 Connection



24 V inputs		
Terminal	**Signal**	**Description**
DI 1	Start [1]	24 V = Start feeding. Continuous signal for several bags or *pulse (min. 200 ms)* for one bag
DI 2	Abort [16]	24 V pulse min. 200 ms. Change to step 1
DI 3	Clear alarm [9]	24 V pulse min. 200 ms. Alarm acknowledgement *(see alarms)*
DI 4	Remote [10]	Switching Local/Remote if the function is enabled according to parameter SYS.RIN 0 V = Local *(nominal value display)* 24 V = Remote *(nominal value external)*
DI 5	Release discharge [4]	24 V pulse min. 200 ms. Release discharge of the scale.
DI 6	FC alarm [11]	24 V = Fault frequency converter when TCON.FIIN = OFF 0 V = Fault frequency converter when TCON.FIIN = ON
DI 7	Low level probe surge bin [12]	24 V = Probe uncovered *(low level)*. Every bag is checked and corrected.
DI 8	Continue feeding [13]	24 V = Continue feeding without zeroing. The signal must be applied together with the start signal at the start. It is therefore possible to finish feeding a bag *(or big bag)* after an interruption.
DI 9 1)	Acknowledgem. tolerance alarms [14]	24 V pulse min. 200 ms. Acknowledgement of tolerance alarms *(continuous signal possible)*
DI 10 1)	Clear bag counter/total [2]	24 V pulse min. 200 ms. The bag counter and the total weight are reset *(only possible in step 1 or 9).*
DI 11 1)	Bed depth adjustment [20]	24 V = enable bed depth adjustment MWSK to setpoint
DI 12 1)	Bed depth closing [21]	24 V = fully close enable bed depth adjustment MWSK

1) On IO extension option
[ ] Signal number *(for reassignment of the inputs/outputs, see SYS.DI/DO)*

24 V outputs		
Terminal	**Signal**	**Description**
DO 1	Alarm [1]	0 V = Alarm if SYS.AINV = ON
DO 2	Weighing ended [2]	24 V = Weighing is completely ended 0 V = At change to step 8
DO 3	Enable bagging [11]	24 V = The bag counter is > 0 at bag preselection or the bag counter nominal value is 0
DO 4	Feeding ended [12]	24 V = Feeding is ended 0 V = At change to step 8
DO 5	Discharge [5]	24 V = Discharge activated
DO 6	Dribble feeding [6]	24 V = Dribble feeding *(also enable frequency converter)*
DO 7	Fast feeding [13]	24 V = Fast feeding
DO 8	Dump pulse [4]	24 V pulse 1 second from start of emptying
DO 9 1)	Step 1 [3]	24 V = The control unit is in step 1
DO 10 1)	Operation [18]	24 V during feeding up to start of emptying
DO 11 1)	Tolerance alarm [9]	24 V = The bag actual weight is outside the set tolerance limits REC.TOL+, REC.TOL–
DO 12 1)	Bed depth Motor OPEN [25]	24 V = motor output increase bed depth *(bed depth adjustment MWSK)*
DO 13 1)	Bed depth Motor CLOSE [26]	24 V = motor output reduce bed depth *(bed depth adjustment MWSK)*
DO 14 1)	Bed depth setpoint=actual [27]	24 V = the bed depth corresponds to the recipe value *(bed depth adjustment MWSK)*
DO 15 1)	Bed depth 0 (CLOSE) [28]	24 V = the bed depth is 0; gate closed *(bed depth adjustment MWSK)*
DO 19 1)	Emptying delayed [19]	24 V = Delayed release signal Emptying. This enables delayed closing of the shutoff flap of the MSDB double-scale system. For delay time see REC.FDEL.
DO 20 1)	Filter cleaning [20]	24 V = Pulse with 0.2 s. Cyclical cleaning of a filter every n-th bag. n can be set at the parameter TCON.CLN.

1) On IO extension option
[ ] Signal number (for reassignment of the inputs/outputs, see SYS.DI/DO)

6.4 Commissioning

Basically the parameters are set at the factory so that the scale is functional.

Check or set the following points:

- The parameter SYS.TYP must be set to **BAG** *(bagging scale)*
  (only adjustable when calibration switch = CAL)
- If the **nominal values** for the scale are set **remotely** then the parameter SYS.REM must be set accordingly *(see chapter 3.7.5)*.
- Setting the **minimum and maximum bag nominal weights** *(parameter TCON.MINB and TCON.MAXB)* *(only adjustable when calibration switch = CAL)*
- If a **frequency converter** is available for the product feed, set TCON.FINV = ON.
- If the **bag** is used as a **weigh hopper**, set TCON.HOP = ON

Note:
A new recipe must be created for every product and bag weight.

Procedure for creating a recipe:

- If available, a recipe of a similar product should be loaded *(similar spec. weight)*. If no recipe is available, you can begin directly at the next point.
  Load recipe: At the LOAD parameter, enter the recipe number of the similar recipe *(the recipe is loaded in the working recipe [0] in this way)*.
- Set the desired **bag weight** at parameter REC.SIZE *(recipe number 0)*.
- If small bag weights *(less than 20 kg)* are required you must ensure that the fast flow feeding time is not too small. To do this, the notes on fast feeding in the chapter «Description» must be observed.
- Set **check cycle** to 1 at parameter REC.CCYC *(recipe number 0)*.
- Start bagging and produce bags until the **actual bag weight** is correct *(MAIN.BWT)* and the actual dribble flow time *(MAIN.DRIB)* corresponds approximately to the nominal dribble flow time *(REC.TDRI)*.
  If a tolerance alarm occurs this is acknowledged with the «Lamp test» key.
- Set the **check cycle** to 10 at parameter REC.CCYC *(recipe number 0)*.
- **Save recipe**: Enter the desired recipe number at parameter SAVE.
  The working recipe (0) is saved in the preselected recipe memory.

6.5 Parameters and alarms

6.5.1 Parameter overview

Operation parameters

Bag counter/act. weight Rec. No./nominal weight
LOAD Load recipe SAVE Save recipe

Parameter groups

MAIN Main group	REC Recipes	TCON Scale type configuration	SYS	
FAST Fast flow time/weight	NR Recipe number	MINB Min. bag weight		
DRIB Dribble flow time/weight	SIZE Bag nominal weight	MAXB Max. bag weight		
FLO Actual flow rate	EWT Empty weight	FINV Frequency converter		
Σ Total weight	TOL+ Tolerance positive	FIIN Input frequency conv.		
BCT Bag counter	TOL– Tolerance negative	HOP Weigh hopper		
BWT Actual bag weight	CCYC Control cycle	TIP Tipping		
HCPO Position bed depth	DOSW Weight feeding	DC Max. dribble flow correction		
HC+– Bed depth adjustment	STRT Fast feeding start.	TDC Control factor dribble flow time		
	TDRI Dribble flow nom. time	CUTC Control factor dribble cut-off weight		
	CUTW Dribble flow cut-off weight	FC Max. fast flow correct.		
	FFLO Fast feeding perform.	TFC Max. fast flow correct.		
	DFLO Dribble feeding performance	TPRO Max. feed time		
	PROT Write protect. recipe.	TFBL Blocking time fast		
	HCSP Pos. setpt. bed depth	TDBL Blocking time dribble		
	FDEL Shutoff gate delayed	TDMA Max. emptying time		
		TDMI Min. emptying time		
		TSTA Max. standstill time		
		CCLR Mode bag counter		
		HCMD Mode bed depth adjustment		
		CLN Filter cleaning interval		

6.5.2 Parameter description

Code	Operation parameters	Input range
	Normal display: Bag counter and current bag actual weight *(bag counter flashes when nominal number is reached)* Enter key: Nominal bag counter «F» key: Recipe number and bag nominal weight	0…99999
LOAD	Load recipe with «F» key or by changing the number *(from the memory to working recipe 0) (only possible in step 1 or 9)*	1…50
SAVE	Save recipe with «F» key or by changing the number *(save working recipe 0 in the memory)* If «NO-SAVE» appears another number has to be selected or the write protection in the recipe has to be cancelled first *(REC.PROT=OFF)*	1…50

Code	Parameter group MAIN *(main group)*	Input range
FAST	Current fast flow time *(left)* and fast flow cut-off weight *(right)* «F» key: Display start value fast flow time or fast flow pre-switch off weight according to REC.DOSW.	
DRIB	Current dribble flow time *(left)* and dribble flow cut-off weight *(right)* «F» key: Display dribble flow nominal time and start value dribble flow cut-off weight.	
FLO	Step number and current actual flow rate in kg/s	
Σ	Erasable total weight in kg 1) «F» key: Clear bag counter/total in «Local» mode and step 1 or 9	
BCT	Bag counter to the total weight «F» key: Bag counter non-erasable	
BWT	Bag actual weight of bag last checked in kg 1)	
HCPO	Actual value *(position)* of bed depth adjustment MWSK in % *(only if HCMD<>OFF).* F and Plus and Minus keys pressed simultaneously: set bed depth to zero. Setting to zero may be performed only if the gate is closed. .	
HC+-	Manual bed depth adjustment MWSK *(only if HCMD<>OFF).* F and Plus keys pressed simultaneously: enlarge bed depth. F and Minus keys pressed simultaneously: reduce bed depth. The outputs are addressed directly. The recipe value is not affected. In general, the principle applies that the adjustment is possible only in steps 1, 7, 9. Can be additionally adjusted with HCMD=AUT only if the gate adjustment is not active.	

1) Unit according to UNIT; decimal point according to DIV

Code	Parameter group REC *(recipes)*	Input range
NR	Recipe number This recipe number is only for viewing or changing recipes. 0 = working recipe.	0…50 [0]
SIZE	Bag nominal weight in kg 3) 1)	MINB…MAXB [25.00]
EWT	Empty weight in kg 3) 2) 1)	10…9999 [2.50]
TOL+	Nominal weight tolerance positive in kg 3) 2) 1)	0…999 [0.10]
TOL–	Nominal weight tolerance negative in kg 3) 2) 1)	0…999 [0.10]
CCYC	Control cycle 1) The empty and full weight is only checked for every nth bag. If the control cycle is 1, every bag is checked.	1…50 [1]
DOSW	Fast feeding according to weight 1) OFF = fast feeding according to time ON = fast feeding according to weight	OFF/ON [ON]
STRT	Fast feeding startup value 2) 1) Cut-off weight or time according to DOSW *(0 = no fast feeding)*	0…9999 [8.00]
TDRI	Dribble flow nominal time in sec 1)	1.0…20.0 [2.8]
CUTW	Dribble flow cut-off weight in kg 3) 2) 1)	0…2000 [0.10]
FFLO	Fast feeding in % 1)	3…99 [80]
DFLO	Dribble feeding in % 1)	3…99 [8]
PROT	Write protection recipe with SAVE 1)	OFF/ON [OFF]
HCSP	Position setpoint for the MWSK bed depth in % 1) *(only if TCON.HCMD <> OFF)*	5…100 [30]
FDEL	Delayed closing of shutoff gate MSDP. 1) Delay time in sec *(see DO19; s used only for MSDB double-scale system)*	0…9.9 [1.0]

1) The recipe number is displayed between the code and the value
2) Not settable greater than SIZE
3) Unit according to UNIT; decimal places according to DIV
[ ] Standard setting

Code	Parameter group TCON *(type specific configuration)*	Input range
MINB	Minimum bag nominal weight in kg 2) 1)	100…50000 [10.00]
MAXB	Maximum bag nominal weight in kg 2) 1)	100…50000 [50.00]
FINV	Frequency converter	OFF/ON [OFF]
FIIN	Input frequency converter OFF = Alarm with 24V at input ON = Alarm with 0V at input	OFF/ON [OFF]
HOP	Bag as weigh hopper OFF = Scale with weigh hopper ON = Bag = Weigh hopper	OFF/ON [OFF]
TIP	Tipping	OFF/ON [OFF]
DC	Maximum dribble flow cut-off weight correct. in kg 2)	0…2000 [0.10]
TDC	Control factor dribble flow time	0.0…0.8 [0.3]
CUTC	Control factor dribble flow cut-off weight	0.0…0.9 [0.3]
FC	Maximum fast flow cut-off weight correction in kg 2)	0…9999 [2.00]
TFC	Maximum fast flow time correction in sec	0.00…2.00 [0.20]
TPRO	Maximum feed time in sec	2…999 [20]
TFBL	Blocking time fast feeding in sec	0.5…9.9 [1.6]
TDBL	Blocking time dribble feeding in sec	0.5…9.9 [1.6]
TDMA	Maximum emptying time in sec «F» key: Display of the actual emptying time	1.0…99.9 [10.0]
TDMI	Minimum emptying time in sec	0.5…9.9 [0.6]
TSTA	Maximum standstill time in sec	1.0…9.9 [5.0]
CCLR	Mode clear bag counter and total weight OFF = Automatic clearing of the bag counter and the total weight after a recipe change at the next start. ON = Bag counter and total weight can only be cleared with signal «Clear bag counter/total» *(possible with key, 24 V input or serially)*	OFF/ON [OFF]
HCMD	Operating mode bed depth adjustment MWSK belt feed OFF = no bed depth adjustment present MAN= bed depth adjustment manual mode adjustment only with parameter MAIN.HC+– AUT = bed depth adjustment automatic mode adjustment with REC.HCPO or corrections with MAIN.HC+– Connection bed depth potentiometer: + 10 V: X9:3 / 0V: X9:7 / tap: X9:2 *(voltage input 0…10 V)* The MEAF jumpers have to be set as follows: X41=A / X42=B Parameter ADC.SENS = UIN *(this is necessary in order that no alarm 10 «AD10V» is generated)*	OFF/MAN/AUT [OFF]
CLN	Filter cleaning interval Filter cleaning every n-th bag during emptying. No cleaning at 0 *(see DO20).*	0…9999 [0]

1) Only adjustable or executable when the calibration switch is in the «CAL» position

2) Unit according to UNIT; decimal places according to DIV

[ ] Standard setting

6.5.3 Alarms

Alarm number	Meaning	Remedy
1…18	**General alarms**	See chapter 3.8 General alarms
20 MOVE	**No standstill** The scale weight is not steady	– Strong vibrations of the scale 3) – Inlet or outlet gate not tight – Air disturbances – Set max. standstill time or standstill tolerances higher *(TCON.TSTA or ADC.MOT)*
21 DTIME	**Emptying time** The scale is not emptied within the max. emptying time	– The discharge gate does not open or only partly 3) – Set the max. discharge time higher *(TCON.TDMA)*
25 FINV	**Frequency converter** The frequency converter reports a fault	– Check frequency converter 3) – TCON.FIIN set incorrectly
26 TOL	**Tolerance** The bag weight is outside the preselected tolerance *(not at alarm output)*	– Irregular product flow 3) and 5) – Aspiration influence on the weigh hopper – Vibration of the scale – Dribble flow rate too high – Dribble flow nominal time too short *(REC.TDRI)* – When fast feeding according to weight: Fast feeding cut-off weight too low *(REC.STRT)* – When fast feeding according to time: Fast flow time too long *(REC.STRT)* – Wrong dribble flow cut-off weight *(REC.CUTW)* – Increase tolerances *(REC.TOL+/TOL–)*
29 PTIME	**Feeding time** The max. feeding time has been exceeded	– No product feed *(empty)* 3) – Fast flow too little *(defective recipe)* – Max. feeding time too short *(TCON.TPRO)*
30 ZERO	**Zero weight check** The current scale weight has too high a deviation from the original zeroing from calibration at the start *(tolerance 4 % of TCON.MAXB)*	– The discharge gate does not open or only partly 3) – Empty weight *(REC.EWT)* set too high
31 POWER	**Power failure** At every start *(switch on mains)* of the control unit	3)
46 HCRT	**Runtime MWSK bed depth adjustment** The setpoint is not reached within 15 secs. During opening and closing. Only in HCMD=AUT operating mode.	– Check of depth adjustment. 3) – If an element is defective, operation is possible in manual mode if necessary *(without alarm).* – The alarm must be reset with «Clear alarm». Then the depth adjustment attempts to reapproach position.

3) Acknowledgement with key or 24 V input «Clear alarm»
5) Acknowledgement with 24 V input «Acknowledgement tolerance alarm»

7 After-sales service

7.1 Stock-keeping of spare parts

Please use the following pages with the appropriate illustrations when ordering spare parts.
Please note that special manufacturing and delivery specifications are frequently applied to Bühler and non-Bühler parts and that the spare parts supplied by Bühler will always conform to the latest technical state of development.

7.2 Address

Bühler AG
Customer Service Grain Processing
CH-9240 Uzwil, Switzerland
Phone: +41 71 955 30 40
Fax: +41 71 955 33 05
service.switzerland@buhlergroup.com

Bühler AG
Customer Service Engineered Products
CH-9240 Uzwil, Switzerland
Phone: +41 71 955 22 44
Fax: +41 71 955 36 60
customerservice@buhlergroup.com

Note:
The information on the identification plate is to be used to identify the device when making any inquiries at Bühler AG.

7.3 Spare parts, scale control unit MEAF

Basic designs MEAF scale control unit incl. housing				**Order number**	
not calibratable *(housing mild steel)*					
115/230 V AC				EKP – 83122 – 820	
24 V DC				EKP – 83122 – 810	
115/230 V AC	USA version			MEAF – 95017 – 810	1)
24 V DC	USA version			MEAF – 95018 – 810	1)
115 V AC	heatable			EKP – 82082 – 830	2)
230 V AC	heatable			EKP – 82082 – 810	2)
115/230 V AC MEAF-**MWBG**		(EBD 1269)	⚠	EKP – 83173 – 810	4)
115/230 V AC MEAF-**MWBG**	USA version	(EBD 1269)	⚠	MEAF – 95026 – 810	1)
not calibratable *(housing stainless steel)*					
115/230 V AC				EKP – 83122 – 880	
24 V DC				EKP – 83122 – 870	
115…230 V AC	USA version			MEAF – 95019 – 810	1)
24 V DC	USA version			MEAF – 95020 – 810	1)
calibratable *(housing mild steel)*					
115/230 V AC				EKP – 83122 – 840	
24 V DC				EKP – 83122 – 830	
115/230 V AC	USA version			MEAF – 95021 – 810	1)
24 V DC	USA version			MEAF – 95022 – 810	1)
115 V AC	heatable			EKP – 82082 – 840	3)
230 V AC	heatable			EKP – 82082 – 820	3)
calibratable *(housing stainless steel)*					
115/230 V AC				EKP – 83122 – 900	
24 V DC				EKP – 83122 – 890	
115/230 V AC	USA version			MEAF – 95023 – 810	1)
24 V DC	USA version			MEAF – 95024 – 810	1)

1) The USA version includes the scale control unit MEAF plus the option «Display cover»
2) Use the heatable version at ambient temperatures of –30…+50 °C
3) Use the heatable version at ambient temperatures of –30…+40 °C
4) ⚠ **Caution!**
The scale control unit MEAF-**MWBG** is equipped with a **EBD 1269** supply
(see also manual Transflowtron II 66161-1-xx-0411).

7.4 Spare parts, options and individual parts

Options	**Order number**	
Option IO extension (*for calibratable versions already installed)*	EKP-83122-91	
Option Profibus DP *(with assembly material and connector(s))*	EKP-83122-93	
Remote control built-in version *(installed in cabinet doors)*	EKP-83122-96	
Remote control table- or wall-mounted version	EKP-83122-97	
DMS connection box with 10 m cable	EKP-84190-81	
DMS connection box without cable	EKP-84190-82	
Individual parts *(spare parts)*		
Basic print EBD 1268 *(incl. EPROM)*	EKP-83121-81	E
Basic print calibratable *(with TCMP values on sticker)*	EKP-85173-81	(E)
Pulse supply 100...240 V AC *(included only in version 115/230 V AC)*	UNE-80206-001	(E)
Mains module EBD 1269 *(only for MWBG version)*	EKP-84169-81	(E)
Print IO extension as replacement	EKP-84170-81	E
Print Profibus DP as replacement	EKP-84183-81	(E)
Profibus plug	UXE-36405-817	(E)
Display print *(without foil)*	EKP-84168-81	E
Display foil *(without print)*	EVP-40034-01	E
Battery for RAM *(clip-on)*	UNE-74921-029	
EPROM with the latest program version	EKP-85118-81	
Fuse F1 *(only for MWBG version, mains module, 315 mA slow-blow.)*	UNE-22002-017	(E)
Fuse F2 *(only for MWBG version, mains module, 2 A slow-blowing)*	UNE-22002-025	(E)
Fuse F3 *(basic print, 1,25 A slow-blowing)*	UNE-22002-023	E

E = spare part recommended for stockkeeping
(E) = spare part recommended for stockkeeping for corresponding version

7.5 Conversion set, supply unit for MEAF

A conversion set is available for the change from the EBD 1269 power supply unit to the new PULSE supply. The set comprise supply unit, hat track, terminals and assembly material.
2 holes have to be drilled for fastening the hat track when making the conversion.

Part number of the conversion set: UVV–00031624

Caution!

The PULSE supply is **not** suitable for **MWBG** use, it is mandatory to use the EBD 1269 supply.

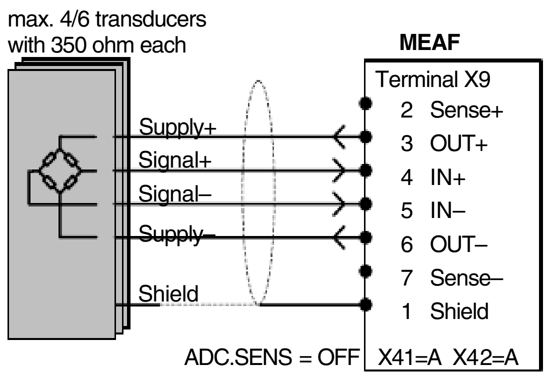
8 Appendix

8.1 Connection of force transducer

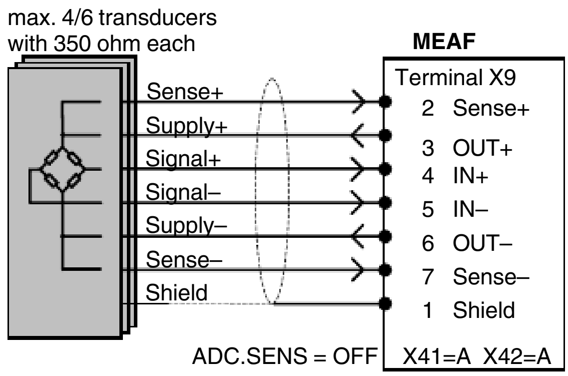
8.1.1 Force transducer types and wire colours

Type / Signal	Reznicek&Hlach DB-10-…	Revere SHBxR-…-C3-SC	HBM Z6…
Sense+		*(yellow)*	green
Supply+	black	green	blue
Signal+	red	white	white
Signal–	white	red	red
Supply–	blue	black	black
~~Sense–~~		~~*(blue)*~~	~~grey~~
Shield	yellow	transparent	yellow

8.1.2 Connection of 4-wire transducer

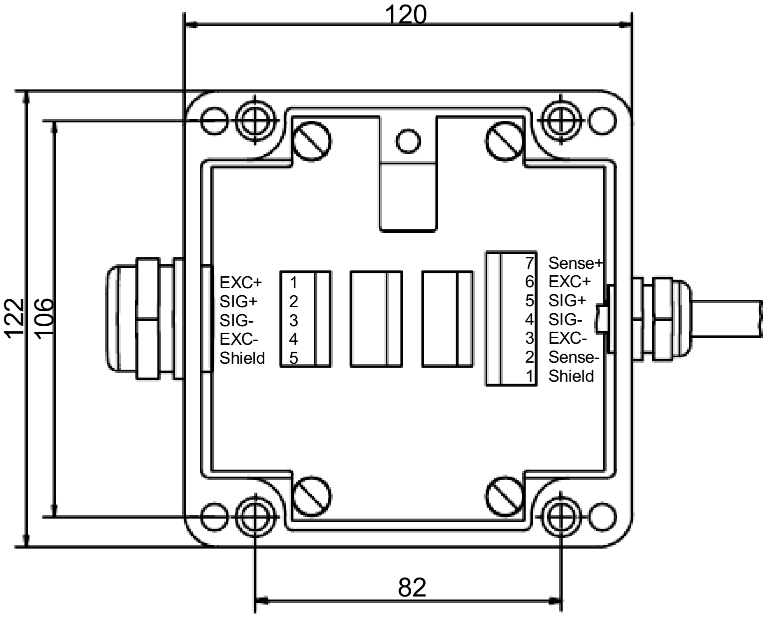


8.1.3 Connection of 6-wire transducer



8.2 DMS connection box

8.2.1 Dimensions and technical data



For versions and order numbers see chapter «7.4»

Material aluminium

Enclosure IP65

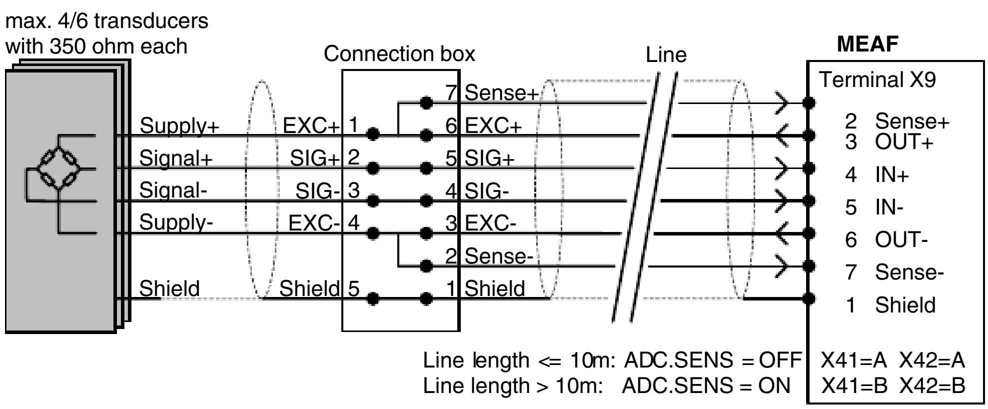
Weight approx. 1 kg

Cable cross section max. 2.5 mm^2

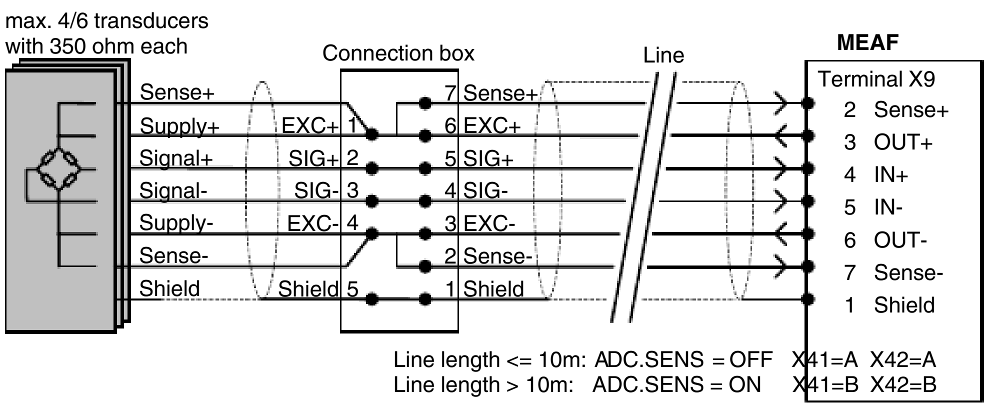
Cover can be sealed

Height: 80 mm

8.2.2 Connection of 4-wire transducer with connection box



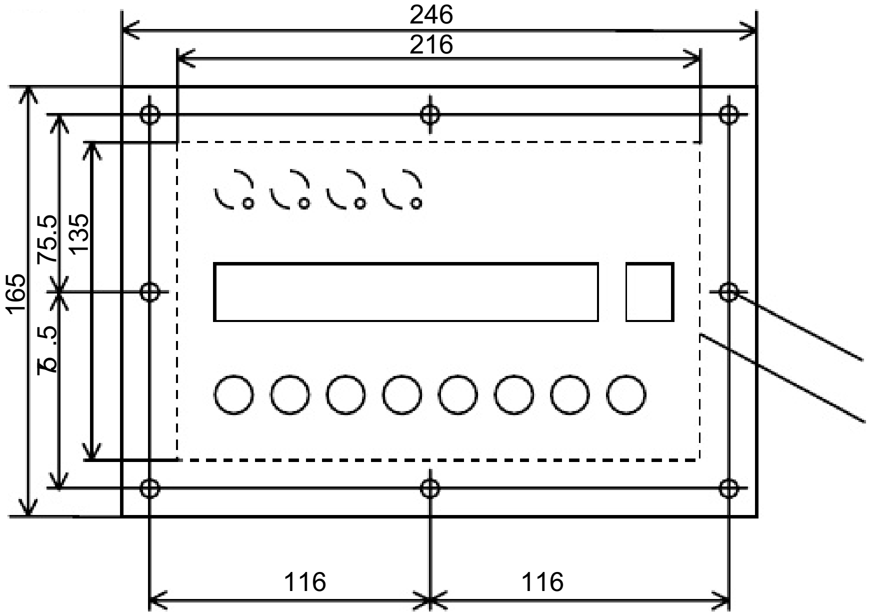
8.2.3 Connection of 6-wire transducer with connection box



8.3 Remote control

In the case of DUMP the key «F7» is assigned with «Start/stop», the key «F8» with «Discharge rest».

8.3.1 Built-in version



For order number see chapter «7.4»

Enclosure foil IP65

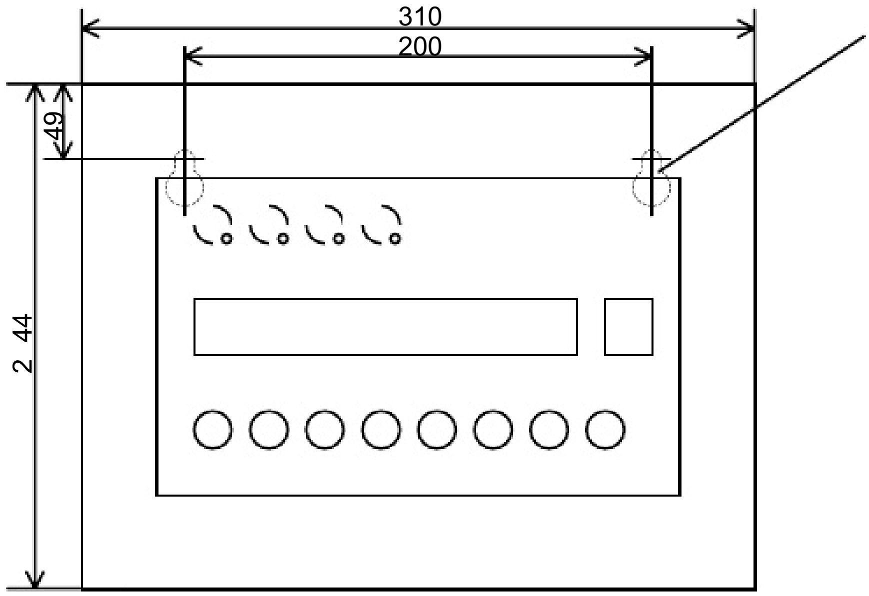
Depth 40 mm

Power supply 24 V DC from MEAF max. 100 mA

8 holes 6 mm

Front cutout

8.3.2 Table/wall-mounted version



2 slots for wall mounting in the back panel

For order number see chapter «7.4»

Enclosure IP65

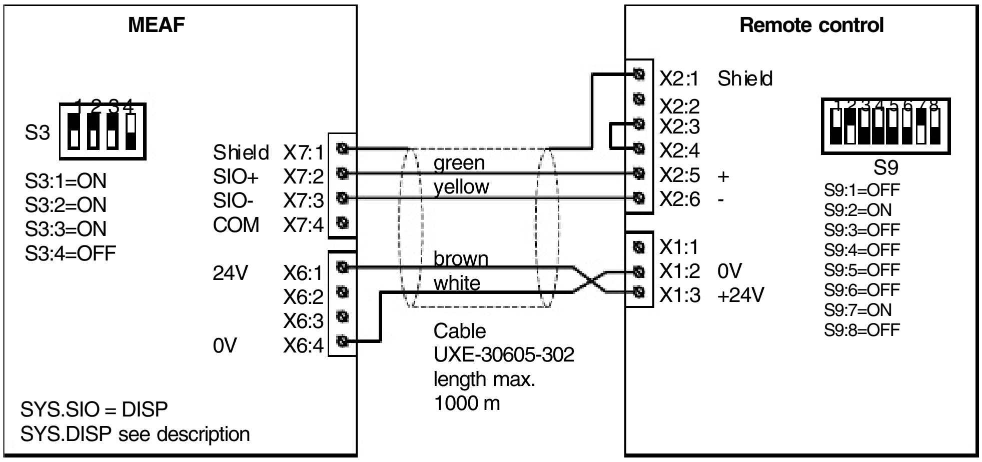
Weight approx. 4 kg

Height rear/top approx. 65 mm

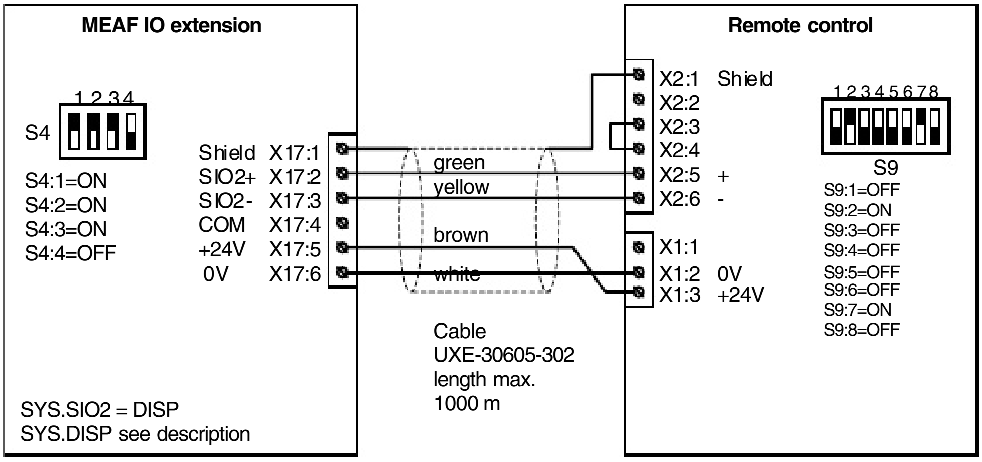
Height front/bottom approx. 45 mm

Power supply 24 V DC from MEAF max. 100 mA

8.3.3 Connection remote control to MEAF basic print *(SIO)*



8.3.4 Connection remote control to MEAF IO extension *(SIO2)*

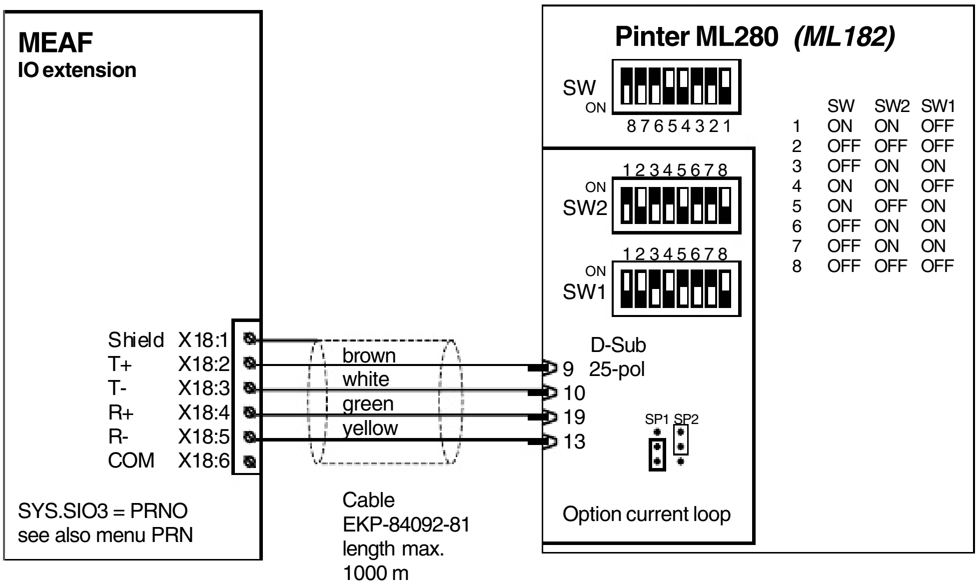


8.4 Printer

For the adjustment of the printer parameters see parameter group PRN.
The printout of the parameter list *(PPAR=PAR)* is possible with every scale type.

8.4.1 Microline ML280

Connection and settings



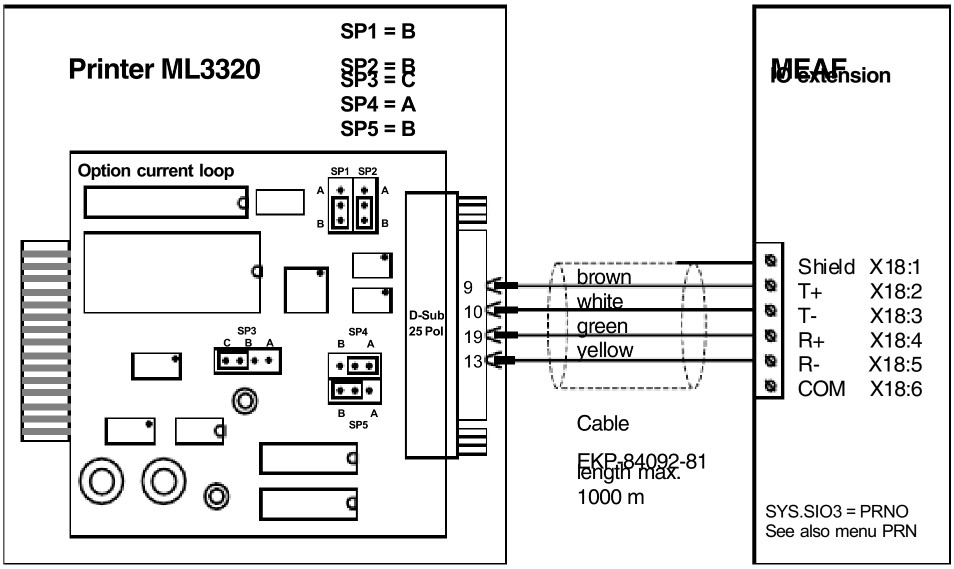
The older model ML182 can also be connected.

2400 baud, 8 bit, even parity
(up to program version 26A 7 Bit SW1:3 = OFF)

Part	Order number
Matrix printer Microline 280 *(ML280)* for 230 V AC	UXE-34951-104
Interface selection Current Loop Auxiliary print must be installed in the printer	UXE-34951-109
Printer cable *(incl. 25-pin plug at printer)* Indicate the length when ordering	EKP-84092-81
Ink ribbon black *(6 pieces)*	UXE-34951-110
Mains transformer 110 V AC for installation into printer Has to be additionally ordered for 110 V AC	UXE-34951-112

8.4.2 Microline ML3320

Connection and settings



Additionally the printer has to be configured via the printer menu, as follows:

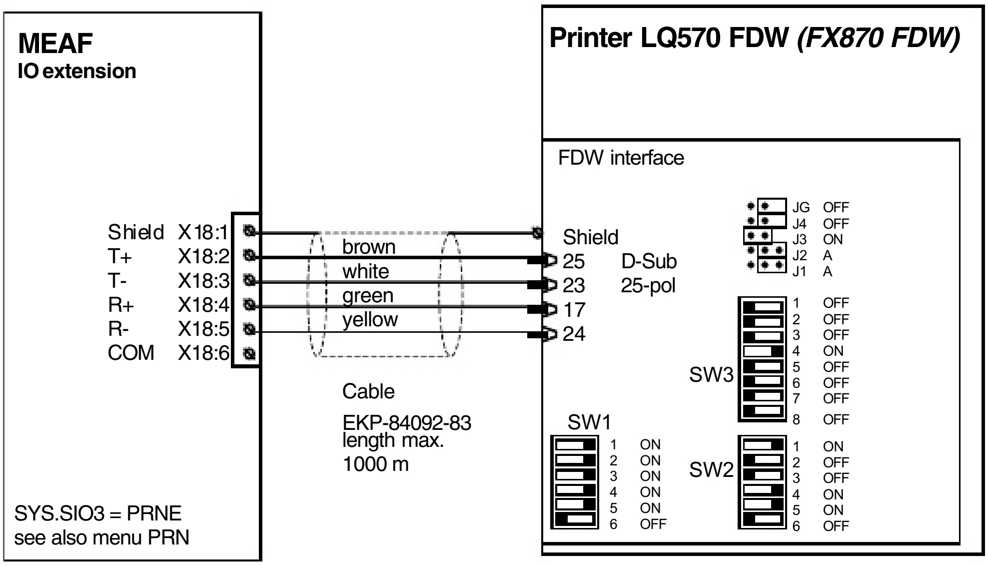
1. Keep the «**SEL**» key + «**LF**» key pressed and at the same time **switch on** the printer **at the main switch**.
   ⇒ All parameters are set back to the ex works setting.
2. Keep the «**SHIFT**» key pressed and at the same time press the «**MENU**» key.
   ⇒ The display lamp **MENU** is lit *(The menu mode is switched on: this means that the functions which are indicated below the keys are valid.)*
3. Press the «**GROUP**» key up to the menu group ***Symbol sets***.
   ⇒ With the «GROUP» key the menu group is changed. The current menu point is printed out in each case. The current menu group can be read on the printed line all the way to the left.
4. Press the «**ITEM**» key up to the menu point ***Code page***.
   ⇒ The «ITEM» key is used to change between the individual menu points, which are available in a menu group. The current menu point can be seen in the centre of the printed line.
5. Press the «**SET**» key until the menu value ***Multilingual*** is set.
   ⇒ The menu value of a certain menu point is edited with the «SET» key. The current menu value can be seen on the right of the printed line. The set of characters Multilingual *(ID 850)* is selected as a standard. For Russian the set of characters Cyrillic 2 *(ID 866)* is selected.
6. Press the «**GROUP**» key up to the menu group ***Serial I/F***.
   ⇒ The first menu point is ***Parity****,* which is adjusted with the next step.
7. Press the «**SET**» key until the menu value ***Even*** is set.
8. Press the «**ITEM**» key up to the menu point ***Baud Rate***.
9. Press the «**SET**» key until the menu value is set to ***2400 BPS***.
10. In order to recheck and document the menu settings, the entire menu setting can be printed out with the «**PRINT**» key.
11. Keep the «**SHIFT**» key pressed and at the same time press the «**MENU**» key in order to exit the menu mode.
    ⇒ The lamp ***SEL*** is lit. The printer is now in the ON-LINE status.
12. Switch the printer off and on again.

Part	Order number
Matrix printer Microline 3320 *(ML3320)* for 230 V AC	UXE-34951-220
Matrix printer Microline 3320 *(ML3320)* for 115 V AC	UXE-34951-210
Interface selection Current Loop Auxiliary print must be installed in the printer	UXE-34951-117
Printer cable *(incl. 25-pin plug at printer)* Indicate the length when ordering.	EKP-84092-81
Single sheet feed for ML3320	UXE-34951-221
Ink ribbon black	UXE-34951-110

8.4.3 EPSON LQ570 FDW

This printer is not for new applications *(replacement ML3320)*.

Connection and settings



The older model EPSON FX870 FDW can also be connected.
1200 baud, 8 bit, even parity
(up to program version 26A 7 Bit SW1:2 = OFF)

Part	Order number
Matrix printer EPSON LQ-570 FDW *(incl. FDW interface)* for 230 V AC	UXE-34951-010
Single sheet feed for LQ-570 FDW	UXE-34951-011
Printer cable *(incl. 25-pin plug at printer)* Indicate the length when ordering	EKP-84092-83
Ink ribbon black	UXE-34951-012

8.4.4 Printing examples

Printing example dump scale *(DUMP)*

```
14-01-99 15:53  Start  SUMS:     0.00  FLOS:  0.000  DWTS: 30.00
     1  30.06      2  30.07      3  29.15      4  30.07      5  30.05
     6  30.06      7  30.01      8  29.96
14-01-99 15:54:23  DCNT     9 FLO 14.054 FWT 30.02 kg EWT  0.01 kg DWT 30.01 kg
14-01-99 15:54:50  Alarm 21 DTIME
14-01-99 15:55:12  DCNT    10 FLO  5.964 FWT 29.96 kg EWT  0.02 kg DWT 29.94 kg
14-01-99 15:55  Stop

14-01-99 16:02
     1                   299.39kg
```

1st line:	Start line with nominal quantity, nominal rate and nominal bulk density at start when PEVT=ON
2nd + 3rd line:	Dump counter and weight of each dumping with PCYC=SHORT
4th + 6th line:	Dump counter, actual rate, full and empty weight and weight of each dumping including time with PCYC=FULL
5th line:	Alarm line when PEVT=ON
7th line:	Stop line at stopping when PEVT=ON
8th line:	Empty line, is inserted before a job printout. A new page will be started if PLEN > 0.
9th + 10th line:	Job printout with date, total number and total with «Clear total» when PJOB=ON. Further settings: PLEN=0, DPOS=1 1, NPOS=2 1, SPOS=2 20

Printing example differential-dosing scale *(DIFF)*

```
25-01-99 11:22  Start  REC: 0  FLOS:  7.000  SUMS:     0.00
  7.002   6.999   7.002   7.001   7.001   7.003   7.001   6.981   6.987   6.997
  6.994   6.996   6.999   6.999   7.004
25-01-99 11:27  Alarm 34 NOPRO
  0.000   0.000   6.774   6.958   6.987
25-01-99 11:28:35  FLO:  7.007  MON:F   0.007  SUM:    555.31  WT: 13.81
25-01-99 11:28:52  FLO:  7.065  MON:F   0.065  SUM:    588.28  WT: 14.09
25-01-99 11:28  Stop
25-01-99 11:29  Job   REC: 0  FLOS:  7.000   SUM:    595.08
```

1st line:	Start line with recipe number, nominal rate und nominal quantity at start when PEVT=ON
2nd + 3rd + 5th line:	Actual rate of each cycle when PCYC=SHORT
4th line:	Alarm line when PEVT=ON
6th + 7th line:	Actual rate, control information, total weight and scale weight with each cycle when PCYC=FULL
8th line:	Stop line at stopping when PEVT=ON
9th line:	Job printout with recipe number, nominal rate and total with «Clear total» when PJOB=ON

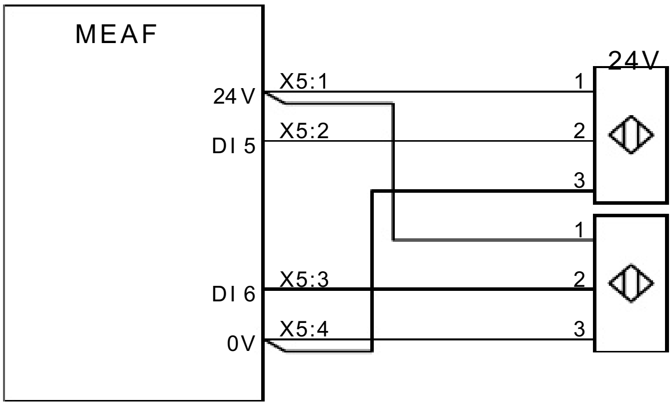
Printing example bagging scale *(BAG)*

```
21-01-99 16:35  Start  REC: 1  SIZE: 25.00  CNT:    0
    1  25.02     2  25.01     3  25.01
21-01-99 16:36  Alarm 29 PTIME
21-01-99 16:40  Start  REC: 1  SIZE: 25.00  CNT:    0
21-01-99 16:41:02  CNT:    4  BWT: 24.99 FAST: 14.50s > 1.98 DRIB: 3.04s  0.19
21-01-99 16:41:23  CNT:    5  BWT: 25.01 FAST: 14.56s > 1.94 DRIB: 2.93s  0.19
21-01-99 16:41:44  CNT:    6  BWT: 24.99 FAST: 14.55s > 1.92 DRIB: 2.90s  0.19
21-01-99 16:42  Job   REC: 1  SIZE: 25.00  CNT:    6  SUM:    150.03
```

1st + 4th line:	Start line with recipe number, nominal weight of bag and number of bags at start when PEVT=ON
2nd line:	Dump counter and bag weight with PCYC=SHORT *(only checked bags)*
3rd line:	Alarm line when PEVT=ON
5th…7th line:	Dump counter, bag weight, fast flow time and weight, dribble flow time and weight with time at PCYC=FULL *(only checked bags)*
8th line:	Job printout with recipe number, nominal weight of bag, number of filled bags and total weight at recipe change when PJOB=ON

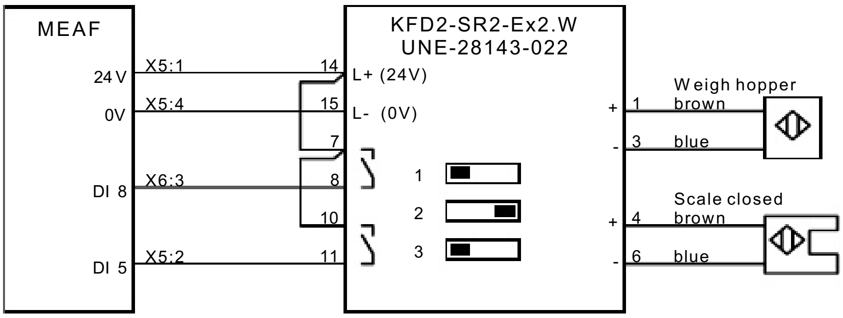
8.5 Connection of probes with dump scales *(DUMP)*

Probes «Scale closed» at the MWBL scale



With 2 probes «Scale closed»: Parameter TCON.CTST = ON1+2

Probe «Scale closed» and probe «Weigh hopper» at MWBB/MWBS scale



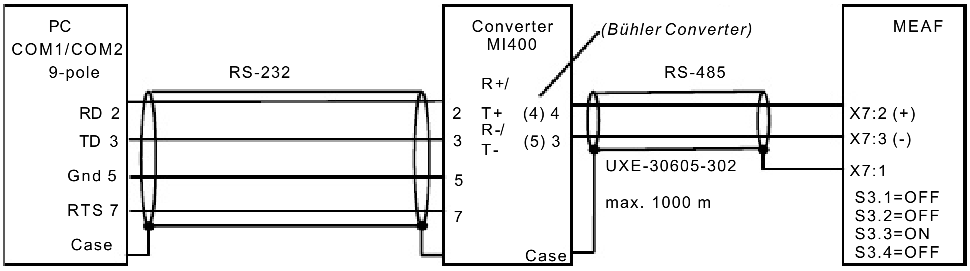
Probes 8 V for explosion proof applications with isolated signal transmitter.
If the probe «Scale closed» is available: Parameter TCON.CTST = ON1
If the probe «Weigh hopper» is available: Parameter TCON.HLPB = ON

8.6 Connection to MWET/MEAF-PC

The MEAF scale control unit can be connected to a PC with the MWET-PC program *(from version 3A)*. With this it is possible to remote control a dump scale *(mainly intake or loading scales)*. Texts according to the customer's requirements can be printed with the «Job printout». A maximum of 6 scales can be connected to the PC and it is possible to mix MEAF and MWET scale control units.

The PC program is described in the MWET manual *(66272)* in the chapters 4.3 to 4.6. The PC interfaces are described in the chapter 8.3. All other chapters in the MWET manual are not used for this application!

Connection



Switch settings at the converter MI400: ***(X = ON)***

		232								485									
DTE	DCE	1	2	3	4	5	6	7	8	1	2	3	4	5	6	7	8	9	10
X			X		X				X					X	X	X	X	X	

Parameter settings at the MEAF:

TCON.HTYP	= ETPC	[1])	SYS.REM	= REMS	[1])	HOST.ADR	= xx	[2])
TCON.STRT	= ON		SYS.EIN	= ON		HOST.BAUD	= 9600	[1])
TCON.DSTO	= ON		SYS.SIO	= HOST	[1])	HOST.WFOR	= FIX	[1])
TCON.KSTO	= ON							

[1]) Setting is compulsory

[2]) Must be set as at the PC *(menu «Configuration.scale»)*

Part	**Order number**
Interface converter MI400 *(RS232 / RS485)* 230 VAC	EKP-85136-81
Interface converter MI400 *(RS232 / RS485)* 120 VAC	EKP-85136-82

Interface converter incl. cable to the PC 1.8 m

8.7 Parameter lists

8.7.1 Parameter list DIFFG / DIFF / DIFFM

Plant: .... Name: .... Date: ....

Parameter	[Default]	Set value
SYS		
TYP	[-----]	= **DIFF…**
REM	[LOC]	=
RIN	[OFF]	=
EIN	[OFF]	=
DISP	[LOC]	=
ACLR	[OFF]	=
AINV	[ON]	=
WIMP	[10.00]	=
IMPM	[TIME]	=
AO	[0MA]	=
AO2	[0MA]	=
AI	[0MA]	=
DI1…	[1,2,9,10,3,11,15	=
DI14	13,0,0,0,0,0,0]	
DO1…	[1,2,19,4,5,6,14,7	=
DO20	3,15,16,17,18,20]	
SIO	[HOST]	=
SIO2	[HOST]	=
SIO3	[PRNO]	=
ADC		
CALW	[40.00]	=
MAXR	[0.00]	=
DIV	[0.01]	=
UNIT	[KG]	=
FILT	[2]	=
MOT	[0.5]	=
SENS	[OFF]	=
AZER	[0.00]	=
NORM	[INDST]	=
TCMP	[100 100]	=
HOST		
ADR	[0]	=
BAUD	[4800]	=
TOUT	[0]	=
WFOR	[FIX]	=
ENBL	[OFF]	=
PRN		
PEVT	[OFF]	=
PCYC	[OFF]	=
PJOB	[OFF]	=
PLEN	[0]	=
LFTM	[0]	=

Parameter	[Default]	Set value
SERV		
VER	[-]	=
ERR	[-]	=
CPU	[-]	=
SADR	[-]	=
PCNT	[-]	=
DMS	[-]	=
+10 V	[-]	=
P-DP	[-]	=
TCON		
MODE	[FLOCON]	=
ADDM	[OFF]	=
EMOD	[OFF]	=
RVA	[ON]	=
VOL	[48]	=
HLEV	[95]	=
LLEV	[35]	=
NLEV	[70]	=
FMAX	[10.000]	=
FMIN	[0.500]	=
MAST	[10.000]	=
CTOL	[5.0]	=
FTOL	[1.0]	=
RCOR	[1.0]	=
TFIL	[25.0]	=
TASP	[0.3]	=
TSTA	[1.2]	=
TMES	[2 60]	=
TFGO	[1.0]	=
TFG	[10]	=
FTYP	[30000]	=
FCOR	[100.0]	=
AOFF	[0]	=
MIX	[50.00]	=
FIIN	[OFF]	=
KSTO	[ON]	=

8.7.2 Recipe list DIFFG / DIFF / DIFFM

Plant:.. Name:.. Date:...................................

NO [0]	DENS [0.75]	FACT [100.0]	CFAC [ON]	CWT [0]	Remarks
0					
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
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42					
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45					
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47					
48					
49					
50					

8.7.3 Parameter list DUMP

Plant: .... Name: .... Date: ....

Parameter	[Default]	Set value
SYS		
TYP	[-----]	= **DUMP**
REM	[LOC]	=
RIN	[OFF]	=
EIN	[OFF]	=
DISP	[LOC]	=
ACLR	[OFF]	=
AINV	[ON]	=
WIMP	[10.00]	=
IMPM	[TIME]	=
AO	[0MA]	=
AO2	[0MA]	=
AI	[0MA]	=
DI1…	[1,2,3,4,5,6,7,8	=
DI14	9,23,24,13,25,16]	
DO1…	[1,2,3,4,5,6,7,8,	=
DO20	9,10,11,0,…,0]	
SIO	[HOST]	=
SIO2	[HOST]	=
SIO3	[PRNO]	=
ADC		
CALW	[40.00]	=
MAXR	[0.00]	=
DIV	[0.01]	=
UNIT	[KG]	=
FILT	[2]	=
MOT	[0.5]	=
SENS	[OFF]	=
AZER	[0.00]	=
NORM	[INDST]	=
TCMP	[100 100]	=
HOST		
ADR	[0]	=
BAUD	[4800]	=
TOUT	[0]	=
WFOR	[FIX]	=
ENBL	[OFF]	=
PRN		
PEVT	[OFF]	=
PCYC	[OFF]	=
PJOB	[OFF]	=
PLEN	[0]	=
LFTM	[0]	=
MJOB	[OFF]	=
PALA	[OFF]	=

Parameter	[Default]	Set value
JOBL	[0]	=
DPOS	[1 1]	=
NPOS	[1 20]	=
SPOS	[1 40]	=
S2PO	[0 1]	=
SERV		
VER	[-]	=
ERR	[-]	=
CPU	[-]	=
SADR	[-]	=
PCNT	[-]	=
DMS	[-]	=
+10V	[-]	=
P-DP	[-]	=
TCON		
MAXW	[50.00]	=
MINW	[10.00]	=
MINT	[20.00]	=
FMAX	[0.000]	=
FAVG	[4]	=
MEWT	[3.00]	=
CTWT	[0.5]	=
CTST	[OFF]	=
IMOD	[OFF]	=
HLPB	[OFF]	=
DIPB	[OFF]	=
INPB	[OFF]	=
PRES	[OFF]	=
FAO	[10.000]	=
ACUT	[OFF]	=
CCYC	[1]	=
TARD	[0]	=
TDMA	[15.0]	=
TDMI	[2.0]	=
TINT	[2.0]	=
TDAP	[0.6]	=
TDAD	[0.4]	=
TASP	[0.5]	=
TBLK	[1.5]	=
TSTA	[5.0]	=
HTYP	[STD]	=
STRT	[OFF]	=
DSTO	[OFF]	=
KSTO	[OFF]	=
FDP	[OFF]	=

8.7.4 Parameter list BAG

Plant:.. Name:.. Date:...................................

Parameter	[Default]	Set value
SYS		
TYP	[-----]	= **BAG**
REM	[LOC]	=
RIN	[OFF]	=
EIN	[OFF]	=
DISP	[LOC]	=
ACLR	[OFF]	=
AINV	[ON]	=
AO	[0MA]	=
AO2	[0MA]	=
AI	[0MA]	=
DI1…	[1,16,9,10,4,11	=
DI14	12,13,14,2,0,..,0]	
DO1…	[1,2,11,12,5,6,13	=
DO20	4,3,18,9,0,…,0]	
SIO	[HOST]	=
SIO2	[HOST]	=
SIO3	[PRNO]	=
ADC		
CALW	[40.00]	=
MAXR	[0.00]	=
DIV	[0.01]	=
UNIT	[KG]	=
FILT	[4]	=
MOT	[0.5]	=
SENS	[OFF]	=
AZER	[0.00]	=
NORM	[INDST]	=
TCMP	[100 100]	=
HOST		
ADR	[0]	=
BAUD	[4800]	=
TOUT	[0]	=
WFOR	[FIX]	=
ENBL	[OFF]	=
PRN		
PEVT	[OFF]	=
PCYC	[OFF]	=
PJOB	[OFF]	=
PLEN	[0]	=
LFTM	[0]	=

Parameter	[Default]	Set value
SERV		
VER	[-]	=
ERR	[-]	=
CPU	[-]	=
SADR	[-]	=
PCNT	[-]	=
DMS	[-]	=
+10V	[-]	=
P-DP	[-]	=
TCON		
MINB	[10.00]	=
MAXB	[50.00]	=
FINV	[OFF]	=
FIIN	[OFF]	=
HOP	[OFF]	=
TIP	[OFF]	=
DC	[0.10]	=
TDC	[0.3]	=
CUTC	[0.3]	=
FC	[2.00]	=
TFC	[0.20]	=
TPRO	[20]	=
TFBL	[1.6]	=
TDBL	[1.6]	=
TDMA	[10.0]	=
TDMI	[0.6]	=
TSTA	[5.0]	=
CCLR	[OFF]	=
HCMD	[OFF]	=
		=
		=
		=
		=
		=
		=
		=
		=
		=
		=
		=

8.7.5 Recipe list BAG

Plant:... Name:.. Date:......................

NR [0]	SIZE [25.00]	EWT [2.50]	TOL+ [0.10]	TOL- [0.10]	CCYC [1]	DOSW [ON]	STRT [8.00]	TDRI [2.8]	CUTW [0.10]	FFLO [80]	DFLO [8]	HCSP [30]	Remarks
0													
1													
2													
3													
4													
5													
6													
7													
8													
9													
10													
11													
12													
13													
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8.7.6 Frequency converter for DIFFM

The parameters are configured at the factory.

If frequency converter ATV is used directly from the manufacturer: enter parameters as follows.

FC1

code	short label	actual value
Menu Set-		
ACC	Acceleration	0.1 s
dEC	Deceleration	0.1 s
LSP	Low Speed	2.0 Hz
HSP	High Speed	100.0 Hz
ItH	I.Mot.Thermal	1.1 A
Ufr	Voltage Boost	40
SLP	Slip. Comp.	70
Menu drC-		
bFr	Std.Motor F.	50 Hz
UnS	Nom.Mot.Volt	230 V
FrS	Nom.Mot.Freq	50.0 Hz
nCr	Nom.MotorCur	0.9 A
nSP	Nom.Mot.Speed	1335
COS	MotorCos.Phi.	0.66
rSC	R.StatorCold	InIt
tUn	Auto Tuning	POn
Uft	SelectTypeU/F	n
tFr	MaxOutputFreq	100.0 Hz
Menu I-O-		
tCt	Type2WireCtrl	trn
CrL3	Low Speed AI3	0.0
CrH3	HighSpeed AI3	20.0
Menu CtL-		
Fr1	Config.Ref.1	AI3
Menu FLt-		
Atr	Auto. Restart	nO
EtF	ExternalFault	LI3
LEt	External fault config	LO
EPL	StpModeExtFlt	nO
EPL ATEX	StpModeExtFlt	YES *)
OPL	In.Phase Loss	YES

*) ATEX ≙ motor with thermal sensor

Restoring factory settings:

ATV31 drC_FCS = IN_I
ATV31 ENT_key ≥ press and hold down 2 sec.
ATV31 Once completed, display changes from FCS to nO

Caution!
Default parameter values are overwritten.

Resetting all parameters to default values for ATV31:
see frequency converter manual

FC2

code	short label	actual value
Menu Set-		
ACC	Acceleration	0.5 s
dEC	Deceleration	0.1 s
LSP	Low Speed	0.0 Hz
HSP	High Speed	90.0 Hz MSDF_R32
HSP	High Speed	50.0 / 60.0 Hz MSDF_R20
ItH	I.Mot.Thermal	2.6 A
SLP	Slip. Comp.	100
SP2	PresetSpeed2	100.0 Hz
SP3	PresetSpeed3	15.0 Hz
SP4	PresetSpeed4	5.0 Hz
CLI	Current Limit	3.4 A
Menu drC-		
bFr	Std.Motor F.	50 Hz
UnS	Nom.Mot.Volt	230 V
FrS	Nom.Mot.Freq	50.0 Hz
nCr	Nom.MotorCur	2.2 A
nSP	Nom.Mot.Speed	930
COS	MotorCos.Phi.	0.64
rSC	R.StatorCold	InIt
tUn	Auto Tuning	Pon
Uft	SelectTypeU/F	N
tFr	MaxOutputFreq	100.0 Hz
Menu I-O-		
tCC	Ctrl.2/3 Wire	2C
Menu CtL-		
LAC	FunAccessLv	L1
Fr1	Config.Ref.1	AI1
Menu FLt-		
Atr	Auto. Restart	nO
EtF	ExternalFault	LI3
LEt	External fault config	LO
EPL	StpModeExtFlt	nO

code	short label	actual value
EPL	StpModeExtFlt	YES *)
OPL	O/PPhaseLoss	YES

*) ATEX ≙ motor with thermal sensor